4. In an electromagnetic wave, the electric field ($\vec{E}$) and the magnetic field ($\vec{B}$) oscillate along x-axis and y-axis respectively. The wave is propagating along :

(A) z-axis (B) – z-axis

(C) x-axis (D) – y-axis

CBSE 2026C

4. In an electromagnetic wave, the electric field ($\vec{E}$) and the magnetic field ($\vec{B}$) oscillate along x-axis and y-axis respectively. The wave is propagating along :

(A) z-axis (B) – z-axis

(C) x-axis (D) – y-axis

CBSE 2026C

(A) z-axis

5. Which of the following is also referred to as 'heat waves' ?

(A) Gamma rays

(B) X-rays

(C) UV rays

(D) Infrared rays

CBSE 2026C

5. Which of the following is also referred to as 'heat waves' ?

(A) Gamma rays

(B) X-rays

(C) UV rays

(D) Infrared rays

CBSE 2026C

(D) Infrared rays

7. If v_X, v_G and v_U are frequencies of X-rays, Gamma-rays and UV rays respectively, then :

(A) $v_X > v_G > v_U$

(B) $v_G > v_X > v_U$

(C) $v_U > v_X > v_G$

(D) $v_X > v_U > v_G$

CBSE 2026C

7. If ν_X, ν_G and ν_U are frequencies of X-rays, Gamma-rays and UV rays respectively, then :

(A) $\nu_X > \nu_G > \nu_U$

(B) $\nu_G > \nu_X > \nu_U$

(C) $\nu_U > \nu_X > \nu_G$

(D) $\nu_X > \nu_U > \nu_G$

CBSE 2026C

(B) $\nu_G > \nu_X > \nu_U$

5. Choose the correct statement : *1*

(A) Visible light has higher frequencies and shorter wavelengths than that of X-rays.

(B) Visible light has lower frequencies and shorter wavelengths than that of X-rays.

(C) Visible light has higher frequencies and longer wavelengths than that of X-rays.

(D) Visible light has lower frequencies and longer wavelengths than that of X-rays.

CBSE 2026C

5. Choose the correct statement : *1*

(A) Visible light has higher frequencies and shorter wavelengths than that of X-rays.

(B) Visible light has lower frequencies and shorter wavelengths than that of X-rays.

(C) Visible light has higher frequencies and longer wavelengths than that of X-rays.

(D) Visible light has lower frequencies and longer wavelengths than that of X-rays.

(D) Visible light has lower frequency and longer wavelength than that of X-rays.

24. (a) Only an accelerated charge can produce an electromagnetic wave. Explain.

(b) What is the difference between conduction current and displacement current ? *3*

CBSE 2026C

24. (a) Only an accelerated charge can produce an electromagnetic wave. Explain.

(b) What is the difference between conduction current and displacement current ? *3*

CBSE 2026C

(a) Charge oscillating (accelerating) with some frequency produces an oscillating electric field in space, which produces oscillating magnetic field, which in turn is the source of oscillating electric field and so on. Thus, oscillating electric and magnetic fields regenerate each other. Electromagnetic wave thus generated propagates through the space.	2
(b) Conduction Current: The current carried by conductors due to flow of charges. Displacement Current: The current due to changing electric field.	½+½

6. The magnetic field in a plane electromagnetic wave travelling in glass ($n = 1{\cdot}5$) is given by

$$B_y = (2 \times 10^{-7}\ \text{T}) \sin(\alpha x + 1{\cdot}5 \times 10^{11}\ t)$$

where x is in metres and t is in seconds. The value of α is :

(A) $0{\cdot}5 \times 10^3\ \text{m}^{-1}$

(B) $6{\cdot}0 \times 10^2\ \text{m}^{-1}$

(C) $7{\cdot}5 \times 10^2\ \text{m}^{-1}$

(D) $1{\cdot}5 \times 10^3\ \text{m}^{-1}$

CBSE 2026

6. The magnetic field in a plane electromagnetic wave travelling in glass (n = 1·5) is given by

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(C) $7{\cdot}5 \times 10^{2}\ m^{-1}$

(D) $1{\cdot}5 \times 10^{3}\ m^{-1}$

CBSE 2026

(C) $7.5 \times 10^{2}\ m^{-1}$

25. What is displacement current (i_d) ? Considering the case of charging of a capacitor, show that $i_d = \varepsilon_0 \frac{d\phi_E}{dt}$. What is the value of i_d for a conductor across which a constant voltage is applied ? CBSE 2026 3

Displacement current arises due to time varying electric field and hence the time varying electric flux in a region.	1
The plates of the capacitor have an area A & a total charge Q, the magnitude of electric field between the plates:	½
$E=\dfrac{Q/A}{\varepsilon_0}$. ....($i$)	
$\phi_e = EA$ ....(ii)	
Substituting (i) in (ii)	½
$\phi_e = \dfrac{Q}{\varepsilon_0}$	
$\dfrac{d\phi_\varepsilon}{dt} = \dfrac{d}{dt}\left(\dfrac{Q}{\varepsilon_0}\right) = \dfrac{1}{\varepsilon_0}\dfrac{dQ}{dt}$	
$\varepsilon_0 \dfrac{d\phi_\varepsilon}{dt} = \dfrac{dQ}{dt}$. ....(iii)	
If charge Q changes with time, the current i_d	
$i_d = \dfrac{dQ}{dt}$	
From eq. (iii) and (iv)	
$i_d = \varepsilon_0 \dfrac{d\phi_e}{dt}$	½
For constant voltage E = constant, so $\phi_e = 0$	½
$\therefore$ i_d=0	

27. Differentiate between conduction current and displacement current. A capacitor is connected across a source providing time-dependent current. Explain how the total current at an instant is the sum of conduction current and displacement current in the circuit at that instant. CBSE 2026 *3*

27. Differentiate between conduction current and displacement current. A capacitor is connected across a source providing time-dependent current. Explain how the total current at an instant is the sum of conduction current and displacement current in the circuit at that instant. CBSE 2026 *3*

Conduction Current	Displacement current	
Current in conductor due to the flow of charges carrier $I_c = \frac{dQ}{dt}$	Current due to time varying electric field and hence electric flux in a region. $I_D = \varepsilon_0 \frac{d\phi_e}{dt}$	½+ ½

Note: Award full marks if the student writes formula only.

Maxwell corrected Ampere circuital law, that the source of magnetic field is not just conduction current due to flow of charges but also time rate of change of Electric field.

Therefore,

$I = I_C + I_D$

$I = \frac{dQ}{dt} + \varepsilon_0 \frac{d\phi_e}{dt}$ 1

Outside the capacitor plates, only conduction current flows,

$I_D = 0$, $I = I_C$ ½

While inside the capacitor,

$I_D = I$, $I_C = 0$ ½

22. What is meant by displacement current ? A capacitor is being charged by a battery. Show that Ampere-Maxwell law justifies continuity and constancy of the current flowing in the circuit. CBSE 2026 *3*

22. What is meant by displacement current ? A capacitor is being charged by a battery. Show that Ampere-Maxwell law justifies continuity and constancy of the current flowing in the circuit. CBSE 2026 *3*

Displacement current: The current due to varying electric field between the plates of capacitor.	1
Ampere-Maxwell Law	
$\oint \vec{B}.\vec{dl} = \mu_0 (I_c + I_d)$	½
Outside the capacitor, $I_d = 0$	½
$\therefore\ I = I_c$	
Inside the capacitor $I_c = 0$	½
$I = I_d = \epsilon_0 \dfrac{d\phi_e}{dt}$	
$= \epsilon_0 \dfrac{d}{dt}\left(\dfrac{Q}{\epsilon_0}\right)$	
$I = \dfrac{dQ}{dt} = I_c$	
Hence, $I_c + I_d$ has the same value at all points of current.	½

7. The ratio of amplitude of electric field to the amplitude of the magnetic field associated with an electromagnetic wave propagating in glass ($n = 1{\cdot}5$) is :

(A) $3 \times 10^8 \text{ ms}^{-1}$

(B) $2 \times 10^8 \text{ ms}^{-1}$

(C) $3{\cdot}3 \times 10^{-9} \text{ ms}^{-1}$

(D) $5 \times 10^{-9} \text{ ms}^{-1}$

CBSE 2026

7. The ratio of amplitude of electric field to the amplitude of the magnetic field associated with an electromagnetic wave propagating in glass ($n = 1{\cdot}5$) is :

(A) $3 \times 10^8\ ms^{-1}$

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(D) $5 \times 10^{-9}\ ms^{-1}$

CBSE 2026

(B) $2\times10^8 ms^{-1}$

25. What are infrared waves ? Why are these waves referred to as 'heat waves' ? Give any two uses of these waves. CBSE 2026 *3*

25. What are infrared waves ? Why are these waves referred to as 'heat waves' ? Give any two uses of these waves. CBSE 2026 *3*

Infrared waves are the waves produced by hot bodies and molecules. **Alternatively:** They are electromagnetic (em) waves whose wavelength lies between 1mm to700nm. They are referred to as heat waves because water molecules present in most materials readily absorb infrared waves. After absorption, their thermal motion increases, that is, they heat up and heat their surroundings.	1 1
Alternatively: When infrared waves pass through a medium, they heat up the medium. **Uses:** 1. Infrared lamps are used in physical therapy. 2. They help in maintaining Earth's average temperature. 3. They are used in satellites for military purpose. 4. They are used in satellites to observe growth of crops. 5. They are used in remote switches. **(Any Two)**	½ + ½

26. What are microwaves ? How are they produced ? Give any two uses of microwaves. $\boxed{\text{CBSE 2026}}$ *3*

26. What are microwaves ? How are they produced ? Give any two uses of microwaves. CBSE 2026 *3*

Microwaves are short wavelength radio waves with frequencies in gigahertz (GHz).	1
These are produced by special vacuum tubes called klystrons, magnetrons and Gunn diodes.	1
Uses: 1. These are used in radar system in aircraft navigation. 2. These can also be used in radar for the speed guns to time fast balls, tennis serves and automobiles. 3. These can also be used in microwave ovens. **[Any two]**	½ + ½

25. What are X-rays ? How are they produced ? Give two uses of X-rays. *3*

CBSE 2026

25. What are X-rays ? How are they produced ? Give two uses of X-rays. *3*

CBSE 2026

The waves of the electromagnetic spectrum whose wavelength lies between 1 nm to 10^{-3} nm.	1
The X-rays are produced by X-ray tubes or when inner shell electrons move from one energy level to a lower energy level. **Alternatively:** Waves produced when a metal target is bombarded by high energy electrons.	1
Uses of X-rays: 1. Used as a diagnostic tool in medicine. 2. For treatment of certain forms of cancer. 3. Scanning luggage at airports. 4. Inspecting materials for structural flaws. **(Any two)**	½ + ½

3. A plane electromagnetic wave travels through a medium and the magnetic field associated with it is given by

$$B = 5 \times 10^{-8} \sin (3 \times 10^{10}\, t - 150\, x)\ T$$

where x is in metres and t is in seconds.

The velocity of the wave is :

(A) $2{\cdot}0 \times 10^{8}\ ms^{-1}$

(B) $4{\cdot}5 \times 10^{7}\ ms^{-1}$

(C) $3{\cdot}5 \times 10^{7}\ ms^{-1}$

(D) $2{\cdot}5 \times 10^{8}\ ms^{-1}$

CBSE 2026

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(C) $3{\cdot}5 \times 10^{7}\ ms^{-1}$

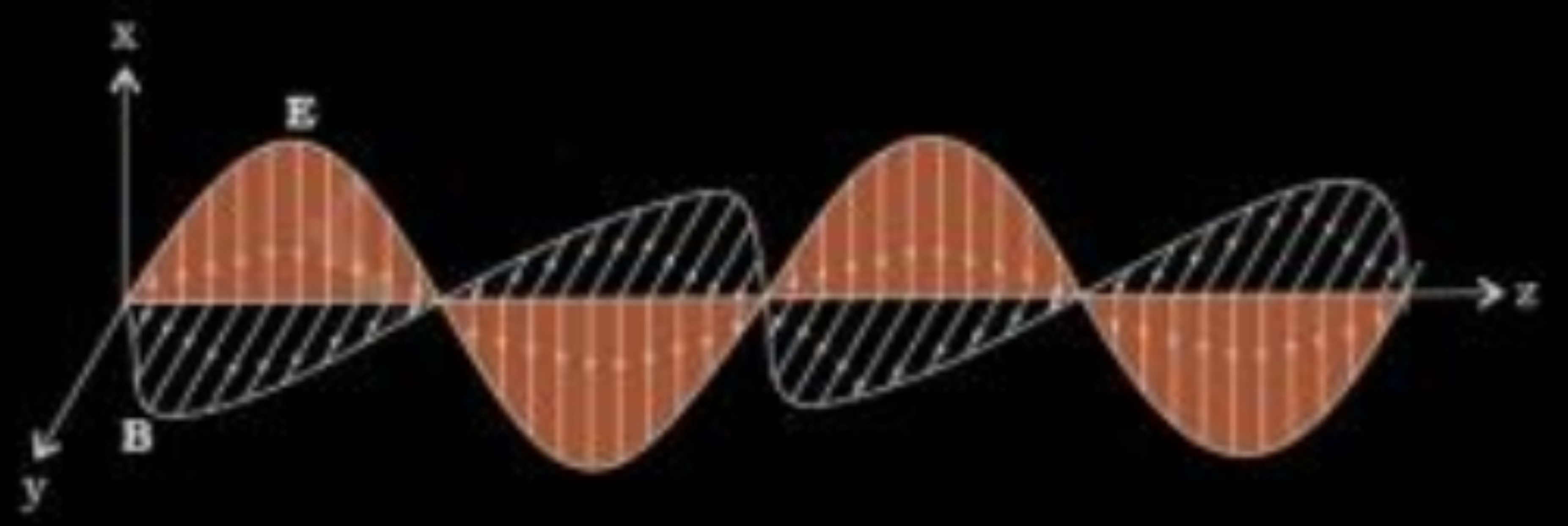
(D) $2{\cdot}5 \times 10^{8}\ ms^{-1}$

CBSE 2026

(A) $2.0 \times 10^{8}\ ms^{-1}$

25. (a) Depict the variation of electric field ($\vec{E}$) and magnetic field ($\vec{B}$) with respect to the direction of propagation of an electromagnetic wave. Write their two important characteristics.

(b) Show that $\dfrac{1}{\sqrt{\varepsilon_0 \mu_0}}$ gives the velocity of an electromagnetic wave in free space. CBSE 2026 3

(a) 	1
Characteristics of Electromagnetic wave: (i) Electric and magnetic fields are perpendicular to each other. (ii) Speed of Electromagnetic wave in free space or vacuum is equal to $3 \times 10^8\ ms^{-1}$. (iii) Electric and magnetic fields are in same phase. (iv) Electromagnetic waves are transverse in nature. (v) Electromagnetic waves can travel in vacuum. **(any two)**	½ +½
(b) $\frac{1}{\sqrt{\mu_0\varepsilon_0}}$	½
$= \frac{1}{\sqrt{4\pi \times 10^{-7} \times \frac{1}{4\pi \times 9 \times 10^{9}}}}$ $= \sqrt{9 \times 10^{16}}$ $= 3 \times 10^{8}$ m/s = Velocity of electromagnetic wave in free space.	½

13. **Assertion (A) :** Displacement current does not have the same physical effect as the conduction current. **1**

Reason (R) : The concept of displacement current is introduced only to satisfy Kirchhoff's rules.

CBSE 2026

13. **Assertion (A) :** Displacement current does not have the same physical effect as the conduction current. **1**

Reason (R) : The concept of displacement current is introduced only to satisfy Kirchhoff's rules.

CBSE 2026

(D) Both Assertion (A) and reason (R) are false

25. (a) A sinusoidal electromagnetic wave of frequency 40 MHz travels in free space in the x-direction. Calculate the wavelength and time period of the wave.

(b) At some point and at some instant the electric field in the above electromagnetic wave has its maximum value of 750 N/C and is along the y-axis. Calculate the magnitude and direction of the magnetic field at this position and time. **3**

CBSE 2026

(a) $\nu = 40MHz$	
wavelength $\lambda = \frac{c}{\nu} = \frac{3\times10^8}{40\times10^6} = 7.5m$	½
Time period, $T = \frac{1}{\nu} = \frac{1}{40\times10^6}$ $= 2.5\times10^{-8}s$	½
(b) $E_0 = 750N/C$	
Magnitude of magnetic field, $B_0 = \frac{E_0}{c}$	½
$= \frac{750}{3\times10^8} = 250\times10^{-8} = 2.5\times10^{-6}T$	½
$\vec{E}$ and $\vec{B}$ are perpendicular to each other and perpendicular to the direction of propagation $\therefore \vec{B}$ is in +z direction	1

11. An electromagnetic wave is propagating along x-axis. At any instant, the phase difference (in radian) between the electric field ($\vec{E}$) and the magnetic field ($\vec{B}$) associated with the wave is 1

(A) zero (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$ (D) π

CBSE 2026

11. An electromagnetic wave is propagating along x-axis. At any instant, the phase difference (in radian) between the electric field ($\vec{E}$) and the magnetic field ($\vec{B}$) associated with the wave is **1**

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(C) $\frac{\pi}{2}$ (D) π

CBSE 2026

(A) Zero

11. An electromagnetic wave is propagating along x-axis. At any instant, the phase difference (in radian) between the electric field ($\vec{E}$) and the magnetic field ($\vec{B}$) associated with the wave is **1**

(A) zero (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$ (D) π

CBSE 2026

(A) Zero

24. Name the electromagnetic waves which are used for **3**

(i) detection of fractures in bones

(ii) physiotherapy

(iii) radar systems

Also write their wavelength range.

CBSE 2026

24. Name the electromagnetic waves which are used for **3**

(i) detection of fractures in bones

(ii) physiotherapy

(iii) radar systems

Also write their wavelength range.

CBSE 2026

(i)	X-rays Wavelength range: 1 nm to 10^{-3} nm	$^1/_2$ $^1/_2$
(ii)	Infrared waves Wavelength range: 1 mm to 700 nm	$^1/_2$ $^1/_2$
(iii)	Microwaves Wavelength range: 0.1 m to 1 mm	$^1/_2$ $^1/_2$

7. The expression of magnetic fields associated with four electromagnetic waves are given below : 1

I. $B_1 = (4 \times 10^{-6}\ \mathrm{T}) \sin [0.7 \times 10^3 x + 1.4 \times 10^{11}\ t]$

II. $B_2 = (2 \times 10^{-7}\ \mathrm{T}) \sin [0.6 \times 10^3 x + 1.5 \times 10^{11}\ t]$

III. $B_3 = (3 \times 10^{-5}\ \mathrm{T}) \sin [0.5 \times 10^3 x + 1.5 \times 10^{11}\ t]$

IV. $B_4 = (5 \times 10^{-4}\ \mathrm{T}) \sin [0.2 \times 10^4 x + 4.8 \times 10^{11}\ t]$

Which wave is travelling in free space ?

(A) I (B) II

(C) III (D) IV

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III. $B_3 = (3 \times 10^{-5}\ T) \sin [0.5 \times 10^3 x + 1.5 \times 10^{11}\ t]$

IV. $B_4 = (5 \times 10^{-4}\ T) \sin [0.2 \times 10^4 x + 4.8 \times 10^{11}\ t]$

Which wave is travelling in free space ?

(A) I (B) II

(C) III (D) IV

CBSE 2026

(C) III

24. Name the electromagnetic waves which are used **3**

(i) as a diagnostic tool in medicine

(ii) in remote switches for TV sets

(iii) in water purifiers

Also write their wavelength range.

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(i) as a diagnostic tool in medicine

(ii) in remote switches for TV sets

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Also write their wavelength range.

CBSE 2026

(i)	X−*rays*	1/2
	Wavelength range: 1nm to $10^{-3} nm$	1/2
(ii)	Infrared waves	1/2
	Wavelength range: 1mm to 700 nm	1/2
(iii)	Ultraviolet waves	1/2
	Wavelength range: 400 nm to 1nm	1/2

7. The electric field and the magnetic field components of an electromagnetic wave travelling in vacuum are represented as **1**

$E = m \sin(\alpha z - \beta t)$

$B = n \sin(\alpha z - \beta t)$

where m, n, α and β are constants characterising the wave. Then

(A) $\frac{m}{n} = c, \frac{\beta}{\alpha} \neq c$

(B) $\frac{n}{m} = \frac{\alpha}{\beta} = c$

(C) $\frac{\beta}{\alpha} = c, \frac{m}{n} \neq c$

(D) $\frac{m}{n} = \frac{\beta}{\alpha} = c$

7. The electric field and the magnetic field components of an electromagnetic wave travelling in vacuum are represented as **1**

$E = m \sin(\alpha z - \beta t)$

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where m, n, α and β are constants characterising the wave. Then

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(B) $\frac{n}{m} = \frac{\alpha}{\beta} = c$

(C) $\frac{\beta}{\alpha} = c, \frac{m}{n} \neq c$

(D) $\frac{m}{n} = \frac{\beta}{\alpha} = c$

CBSE 2026

(D) $\frac{m}{n} = \frac{\beta}{\alpha} = c$

27. Name the electromagnetic waves which are (i) also known as 'heat waves', (ii) used in LASIK eye surgery, (iii) used to destroy cancer cells. Also write their wavelength range. CBSE 2026 **3**

27. Name the electromagnetic waves which are (i) also known as 'heat waves', (ii) used in LASIK eye surgery, (iii) used to destroy cancer cells. Also write their wavelength range. CBSE 2026 **3**

(i)	Infrared waves	$^1/_2$
	Wavelength range: 1mm to 700 nm	$^1/_2$
(ii)	Ultraviolet rays	$^1/_2$
	Wavelength range: 400 nm to 1nm	$^1/_2$
(iii)	Gamma rays	$^1/_2$
	Wavelength range: $<10^{-3}$nm	$^1/_2$

7. Electromagnetic waves used in a diagnostic tool in medicine have a wavelength range **1**

(A) 1 nm to 10^{-3} nm

(B) 400 nm to 1 nm

(C) 1 mm to 700 nm

(D) 0.1 m to 1 mm

CBSE 2026

7. Electromagnetic waves used in a diagnostic tool in medicine have a wavelength range 1

(A) 1 nm to 10^{-3} nm (B) 400 nm to 1 nm

(C) 1 mm to 700 nm (D) 0.1 m to 1 mm

CBSE 2026

(A) 1nm to 10^{-3}nm

7. Electromagnetic waves used for purification of water are **1**

(A) X-rays (B) Ultraviolet rays

(C) Infrared rays (D) Ultrasonic rays

CBSE 2026

7. Electromagnetic waves used for purification of water are **1**

(A) X-rays (B) Ultraviolet rays

(C) Infrared rays (D) Ultrasonic rays

CBSE 2026

(B) Ultraviolet rays

11. An electromagnetic wave passes from vacuum into a dielectric medium with relative electrical permittivity (3/2) and relative magnetic permeability (8/3). Then, its **1**

(A) wavelength is doubled and frequency remains unchanged.

(B) wavelength is doubled and frequency is halved.

(C) wavelength is halved and frequency remains unchanged.

(D) wavelength and frequency both will remain unchanged.

11. An electromagnetic wave passes from vacuum into a dielectric medium with relative electrical permittivity (3/2) and relative magnetic permeability (8/3). Then, its **1**

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(C) wavelength is halved and frequency remains unchanged.

(D) wavelength and frequency both will remain unchanged.

CBSE 2026

(C) wavelength is halved and frequency remains unchanged

9. Welders wear special glass goggles or face masks with glass windows to protect their eyes from **1**

(A) Infrared rays (B) Ultraviolet rays

(C) X-rays (D) Microwaves

CBSE 2026

9. Welders wear special glass goggles or face masks with glass windows to protect their eyes from **1**

(A) Infrared rays

(B) Ultraviolet rays

(C) X-rays

(D) Microwaves

CBSE 2026

(B) Ultraviolet rays

6. Which of the following electromagnetic waves are known as 'heat waves' ?

(A) UV rays

(B) IR rays

(C) X-rays

(D) Gamma rays

CBSE 2025 C

6. Which of the following electromagnetic waves are known as 'heat waves' ?

(A) UV rays

(B) IR rays

(C) X-rays

(D) Gamma rays

CBSE 2025 C

(B) IR rays

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CBSE 2025 C

6. Welders wear special glass goggles or face masks with glass windows to protect their eyes from :

(A) UV rays

(B) X-rays

(C) IR rays

(D) Gamma rays

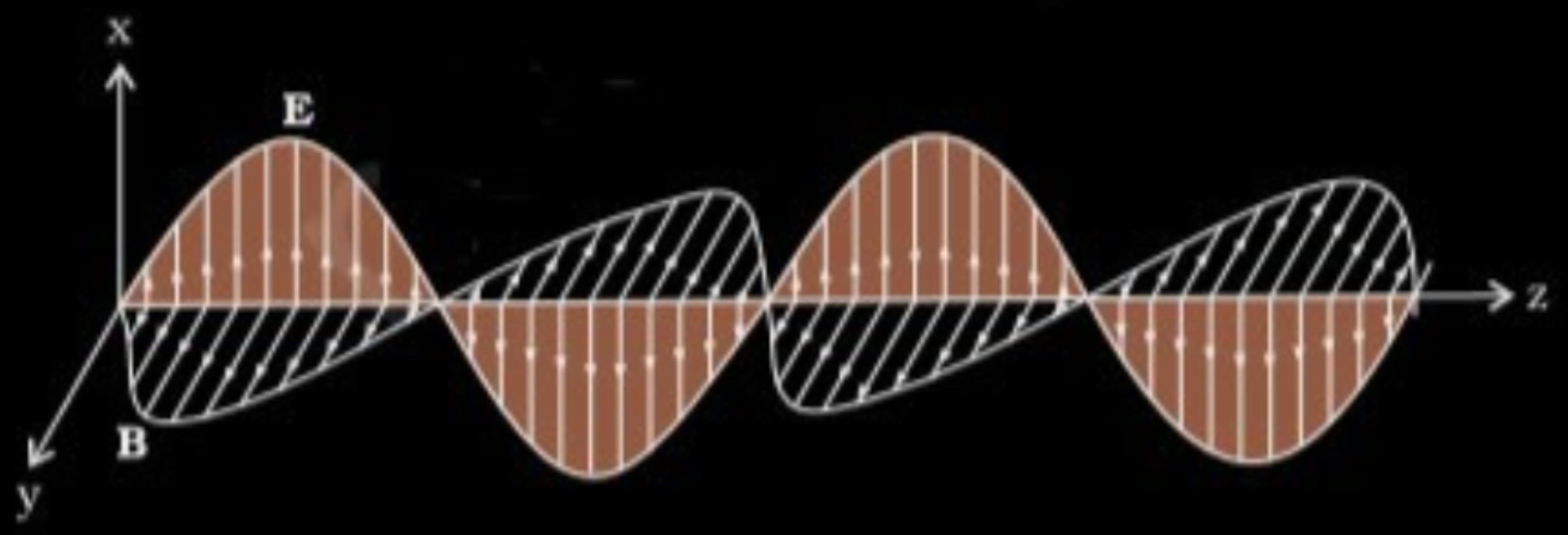
CBSE 2025 C

(A) UV rays

24. Depict the variation of electric and magnetic fields in an electromagnetic wave as it propagates along z-axis. In a plane electromagnetic wave in free space, the electric field oscillates sinusoidally at a frequency of $1 \cdot 5 \times 10^{10}$ Hz with amplitude 36 Vm^{-1}. Find :

(a) the wavelength of the wave, and

(b) the amplitude of the associated magnetic field. CBSE 2025 C *3*

a) 	1
$\lambda = \dfrac{c}{v}$	½
$= \dfrac{3\times10^{8}}{1.5\times10^{10}} = 2\times10^{-2} = 0.02\,\text{m}$	½
b) $B_0 = \dfrac{E_0}{c}$	½
$= \dfrac{36}{3\times10^{8}}$	
$= 1.2\times10^{-7}\ \text{T}$	½

6. Electromagnetic radiation used to kill germs in water purifiers is :

(A) Microwaves
(B) Ultraviolet rays
(C) Gamma rays
(D) Radio waves

CBSE 2025 C

6. Electromagnetic radiation used to kill germs in water purifiers is :

(A) Microwaves (B) Ultraviolet rays

(C) Gamma rays (D) Radio waves

CBSE 2025 C

(B) Ultraviolet rays

7. In free space, which of the following quantities is always larger for X-rays than for a radio wave ?

(A) Speed (B) Frequency

(C) Amplitude (D) Wavelength

CBSE 2025 C

7. In free space, which of the following quantities is always larger for X-rays than for a radio wave ?

(A) Speed (B) Frequency

(C) Amplitude (D) Wavelength

CBSE 2025 C

(B) Frequency

24. How are electromagnetic waves produced ? Write any two characteristics of electromagnetic waves. Give one use each of :

(a) radio waves

(b) microwaves. CBSE 2025 C *3*

24. How are electromagnetic waves produced ? Write any two characteristics of electromagnetic waves. Give one use each of :

(a) radio waves

(b) microwaves. CBSE 2025 C *3*

EMW are produced by accelerating/ oscillating charges	1
Characteristics:	
1. Travel with speed of light in free space.	
2. Transverse in nature.	
3. Exert pressure on the surface they fall.	
4. Carry momentum and energy	½+½
(Or any other two characteristics)	
Uses:	
(i) Radio waves are used for Radio or T.V. communication.	½
(ii) Microwaves: Radar / Satellite communication	½
(Any other use)	

13. *Assertion (A) :* EM waves do not require a medium for their propagation.

Reason (R) : EM waves are transverse waves.

CBSE 2025

13. *Assertion (A) :* EM waves do not require a medium for their propagation.

Reason (R) : EM waves are transverse waves.

CBSE 2025

(B) Both Assertion(A) and Reason (R) are true but Reason(R) is **not** the correct explanation of Assertion(A).

26. Name the electromagnetic radiations and write their frequency range : *3*

(a) Used to take the photograph of the bones

(b) Produced by hot bodies

(c) Used in television communication systems

CBSE 2025

26. Name the electromagnetic radiations and write their frequency range : *3*

(a) Used to take the photograph of the bones

(b) Produced by hot bodies

(c) Used in television communication systems

CBSE 2025

(a) **Name**: X- rays	½
Frequency Range: 10^{16} - 10^{20} Hz	½
(b) **Name**: Infra-red radiation	½
Frequency Range: $10^{12} - 10^{14}$ Hz	½
(c) **Name**: Radio waves	½
Frequency Range: $10^{5} - 10^{9}$ Hz	½

25. (a) How are electromagnetic waves produced ? CBSE 2025

(b) Write the wavelength range and one use of :

(i) Microwaves, and

(ii) Ultraviolet waves. *3*

25. (a) How are electromagnetic waves produced ?

(b) Write the wavelength range and one use of :

(i) Microwaves, and

(ii) Ultraviolet waves. *3*

(a) Electromagnetic waves are produced by an oscillating or accelerated charge. **Alternatively**: An oscillating charge produces an oscillating electric field which produces an oscillating magnetic field which in turn is a source of oscillating electric field & so on.	1
(b) (i) **Microwaves**	½
Wavelength: 0.1 m to 1 mm	
Use : Radar used in aircraft navigation. Microwave ovens. **(Any one)**	½
(ii) **Ultraviolet waves**	½
Wavelength: 400 nm to 1 nm	
Use : Kill germs in UV purifiers. LASIK eye surgery. **(Any one)**	½

25. (a) Differentiate between 'conduction current' and 'displacement current', giving one similarity and one dissimilarity between them.

(b) Explain the existence of electromagnetic waves in free space, using the concept of displacement current. CBSE 2025 *3*

25. (a) Differentiate between 'conduction current' and 'displacement current', giving one similarity and one dissimilarity between them.

(b) Explain the existence of electromagnetic waves in free space, using the concept of displacement current. CBSE 2025 *3*

(a) **Similarity** Both give rise to magnetic field.	1
Dissimilarity(any one) • Conduction current is due to flow of charges in the conductor. • Displacement current arises due to change in electric field/ time varying electric flux.	1
(b) A magnetic field, changing with time, gives rise to an electric field. Then, an electric field changing with time gives rise to a magnetic field and is a consequence of the displacement current being a source of a magnetic field. Thus, time- dependent electric and magnetic fields give rise to each other and hence em wave is generated. **(Note: Please award ½ mark of this part if a child writes the expression only)** $\oint \vec{B}.\vec{dl} = \mu_0 I_c + \mu_0 \varepsilon_0 \frac{d\Phi_e}{dt}$	1

13. *Assertion (A) :* X-rays are produced when slow moving electrons are stopped by a metal target of high atomic number.

Reason (R) : X-rays consist of low-energy photons.

CBSE 2025

13. *Assertion (A) :* X-rays are produced when slow moving electrons are stopped by a metal target of high atomic number.

Reason (R) : X-rays consist of low-energy photons.

CBSE 2025

(D) Both Assertion (A) and Reason (R) are false.

22. Name the electromagnetic wave used (i) in radar, (ii) in eye surgery and (iii) as diagnostic tool in medicine. Write their wavelength range also. *3*

CBSE 2025

22. Name the electromagnetic wave used (i) in radar, (ii) in eye surgery and (iii) as diagnostic tool in medicine. Write their wavelength range also. *3*

CBSE 2025

The electromagnetic waves used are	
(i) Microwaves	½
(ii) Ultraviolet / Infrared	½
(iii) X-rays	½
Wavelength range of electromagnetic waves used	
(i) 0.1 m to 1 mm	½
(ii) 400 nm to 1 nm / 1mm to 700 nm	½
(iii) 1 nm to 10^{-3} nm	½

7. The electromagnetic radiation used to kill germs in water purifiers is **1**

(A) Ultraviolet rays (B) Infrared waves

(C) Visible rays (D) γ-rays

CBSE 2025

7. The electromagnetic radiation used to kill germs in water purifiers is **1**

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CBSE 2025

(A) Ultraviolet rays

26. (a) Write two characteristics of electromagnetic waves. **3**

(b) What is meant by 'displacement current' ? Explain briefly how is displacement current set up during charging of a capacitor in a circuit.

CBSE 2025

26. (a) Write two characteristics of electromagnetic waves. **3**

(b) What is meant by 'displacement current' ? Explain briefly how is displacement current set up during charging of a capacitor in a circuit.

CBSE 2025

(a) Characteristics of em waves(any two) (i) Electromagnetic waves carries energy and momentum. (ii) velocity of electromagnetic wave is same as velocity of light in vaccum. (iii) In electromagnetic wave electric field vector $\vec{E}$, magnetic field vector $\vec{B}$ and direction of propagation, all are mutually perpendicular.	1
(b) Displacement current – It is defined as the current due to changing electric field and hence electric flux with time. **Alternatively** $I_d = \epsilon_o \left(\frac{d\phi_E}{dt}\right)$	1
During charging of capacitor, time varying electric field/ electric flux between the plates of capacitor induces the displacement current.	1

8. Which of the following rays coming from the Sun plays an important role in maintaining the Earth's warmth ?

(A) Infrared rays

(B) γ rays

(C) UV rays

(D) Visible light rays

CBSE 2025

8. Which of the following rays coming from the Sun plays an important role in maintaining the Earth's warmth ?

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CBSE 2025

(A) Infrared Rays

11. The dimensions of $(\mu\varepsilon)^{-1}$, where ε is permittivity and μ is permeability of a medium, are :

(A) $[M^0 L^1 T^{-1}]$

(B) $[M^0 L^2 T^{-2}]$

(C) $[M^1 L^2 T^{-2}]$

(D) $[M^1 L^{-1} T^1]$

CBSE 2025

11. The dimensions of $(\mu\varepsilon)^{-1}$, where ε is permittivity and μ is permeability of a medium, are :

(A) $[M^0 L^1 T^{-1}]$

(B) $[M^0 L^2 T^{-2}]$

(C) $[M^1 L^2 T^{-2}]$

(D) $[M^1 L^{-1} T^1]$

CBSE 2025

(B) $\left[M^0 L^2 T^{-2}\right]$

10. Which of the following electromagnetic waves has photons of largest momentum ?

(A) X-rays

(B) AM radio waves

(C) Microwaves

(D) TV waves

CBSE 2025

10. Which of the following electromagnetic waves has photons of largest momentum ?

(A) X-rays (B) AM radio waves

(C) Microwaves (D) TV waves

CBSE 2025

(A) X- rays

7. In an electromagnetic wave travelling in free space, the amplitude of magnetic field is $6{\cdot}0 \times 10^{-4}$ T. The amplitude of its electric field is :

(A) 2×10^{4} Vm^{-1} (B) $1{\cdot}5 \times 10^{12}$ Vm^{-1}

(C) $1{\cdot}8 \times 10^{5}$ Vm^{-1} (D) $0{\cdot}3 \times 10^{4}$ Vm^{-1}

CBSE 2025

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(C) $1{\cdot}8 \times 10^{5}$ Vm^{-1}

(D) $0{\cdot}3 \times 10^{4}$ Vm^{-1}

CBSE 2025

(C) 1.8×10^{5} Vm^{-1}

28. (a) State any three characteristics of electromagnetic waves.

(b) Briefly explain how and where the displacement current exists during the charging of a capacitor. CBSE 2025 *3*

28. (a) State any three characteristics of electromagnetic waves.

(b) Briefly explain how and where the displacement current exists during the charging of a capacitor. CBSE 2025 *3*

(a) (Any three)	
• Electromagnetic wave carries energy.	½
• Electromagnetic wave carries momentum.	½
• Electromagnetic wave moves with velocity of light in vacuum.	½
• In electromagnetic wave, electric field vector, magnetic field vector and direction of propagation, all are mutually perpendicular.	
• Electromagnetic waves are transverse in nature.	
• Electromagnetic waves do not require a physical medium to propagate and can travel through a vacuum.	
• Electromagnetic waves consist of oscillating electric and magnetic fields.	
(b)	
• During charging of capacitor, time varying electric field / electric flux between the plates of capacitor induces the displacement current.	1
• Displacement current exists between the plates of capacitor.	½

7. The electric field in space between the plates of a parallel plate capacitor (each of area $2 \cdot 5 \times 10^{-3}$ m^2) is changing at the rate of 4×10^6 Vm^{-1}s^{-1}. The displacement current between the plates of the capacitor is :

(A) $1 \cdot 8 \times 10^{-5}$ A (B) $3 \cdot 47 \times 10^{-6}$ A

(C) $8 \cdot 85 \times 10^{-8}$ A (D) $6 \cdot 32 \times 10^{-4}$ A

CBSE 2025

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(C) $8 \cdot 85 \times 10^{-8}$ A (D) $6 \cdot 32 \times 10^{-4}$ A

CBSE 2025

(C) 8.85×10^{-8} A

7. The amplitude of electric field in an electromagnetic wave in free space is 1000 Vm^{-1}. The amplitude of the magnetic field in this electromagnetic wave is :

(A) $3{\cdot}0 \times 10^{-3}$ T

(B) $3{\cdot}33 \times 10^{-8}$ T

(C) $3{\cdot}0 \times 10^{11}$ T

(D) $3{\cdot}33 \times 10^{-6}$ T

CBSE 2025

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(C) $3{\cdot}0 \times 10^{11}$ T

(D) $3{\cdot}33 \times 10^{-6}$ T

CBSE 2025

(D) 3.33×10^{-6} T

14. **Assertion (A)** : Out of Infrared and radio waves, the radio waves show more diffraction effect. **1**

Reason (R) : Radio waves have greater frequency than infrared waves.

CBSE 2025

14. **Assertion (A)** : Out of Infrared and radio waves, the radio waves show more diffraction effect. 1

Reason (R) : Radio waves have greater frequency than infrared waves.

CBSE 2025

(C) If Assertion (A) is true but Reason (R) is false.

25. (a) How is an electromagnetic wave produced ? **3**

(b) An electromagnetic wave is travelling in vertically upward direction. At an instant, its electric field vector points in west direction. In which direction does the magnetic field vector point at that instant ?

(c) Estimate the ratio of shortest wave length of radio waves to the longest wave length of gamma waves. CBSE 2025

25. (a) How is an electromagnetic wave produced ? **3**

(b) An electromagnetic wave is travelling in vertically upward direction. At an instant, its electric field vector points in west direction. In which direction does the magnetic field vector point at that instant ?

(c) Estimate the ratio of shortest wave length of radio waves to the longest wave length of gamma waves.

CBSE 2025

(a) Electromagnetic waves are produced by accelerating / oscillating charges.	1
(b) South direction	1
(c) $\dfrac{\text{Shortest wavelength of radio waves}}{\text{Longest wavelength of gamma waves}} = \dfrac{0.1}{10^{-12}} = 10^{11}$	1

25. (a) The electric field $\vec{E}$ of an electromagnetic wave propagating in north direction is oscillating in up and down direction. Describe the direction of magnetic field $\vec{B}$ of the wave. **3**

(b) Are the wave length of radio waves and microwaves longer or shorter than those detectable by human eyes ?

(c) Write main use of each of the following in human life :

(i) Infrared waves (ii) Gamma rays

CBSE 2025

25. (a) The electric field $\vec{E}$ of an electromagnetic wave propagating in north direction is oscillating in up and down direction. Describe the direction of magnetic field $\vec{B}$ of the wave. **3**

(b) Are the wave length of radio waves and microwaves longer or shorter than those detectable by human eyes ?

(c) Write main use of each of the following in human life :

(i) Infrared waves (ii) Gamma rays

CBSE 2025

(a) Magnetic field oscillates in the east-west direction as it is mutually perpendicular to the direction of electric field.	1
(b) Wavelength of the radiowaves and microwaves is longer than visible light.	1
(c) (i) Use of infrared waves: - (Any one) - Used for physical therapy. - Used for maintaining average temperature of the earth. - Used in Earth satellites for military purposes & to observe growth of crops. - Used in remote control	½
(ii)Use of Gamma rays: - (Any one) - Used in treatment of cancer. - Used in diagnostic imaging. - Used to sterilize medical equipment	½

7. X-rays are more harmful to human beings than ultraviolet radiations because X-rays - **1**

(A) have frequency lower than that of ultraviolet radiations.

(B) have wavelength smaller than that of ultraviolet radiations.

(C) move faster than ultraviolet radiations in air.

(D) are mechanical waves but ultraviolet radiations are electro-magnetic waves.

CBSE 2025

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CBSE 2025

(B) have wavelength smaller than that of ultraviolet radiation

25. (a) A parallel plate capacitor is charged by an ac source. Show that the sum of conduction current (I_c) and the displacement current (I_d) has the same value at all points of the circuit. **3**

(b) In case (a) above, is Kirchhoff's first rule (junction rule) valid at each plate of the capacitor ? Explain.

CBSE 2025

a) $\because$ Total current $I = I_c + I_d$

outside the capacitor

$I_d = 0$ ½

$\therefore I = I_c$

Inside the capacitor

$I_c = 0$ ½

$$\therefore \; I = I_d = \varepsilon_0 \frac{d\phi_E}{dt}$$ ½

$$= \varepsilon_0 \frac{d}{dt}[EA]$$ ½

$$= \varepsilon_0 \frac{d}{dt}\left[\frac{\sigma}{\varepsilon_0} A\right]$$

$$= \frac{\varepsilon_0}{\varepsilon_0} A \frac{d}{dt}\left[\frac{Q}{A}\right]$$

$$I = \frac{dQ}{dt} = I_c$$ ½

Alternatively

$\because$ Total current $I = I_c + I_d$

outside the capacitor

$I_d = 0$ ½

$\therefore I = I_c$

Inside the capacitor

$I_c = 0$ ½

$$I = I_d = \varepsilon_0 \frac{d\phi_E}{dt}$$ ½

$$= \varepsilon_0 \frac{d}{dt}\left[\frac{Q}{\varepsilon_0}\right]$$

$$I = \frac{dQ}{dt} = I_c$$ ½

hence $I_c + I_d$ has the same value at all points of the circuit.

b) Yes 1

Current entering the capacitor is (I_c) and between the plates capacitor is (I_d)

$I_c = I_d$

which validates Kirchhoff's junction rule.

7. Which one of the following correctly represents the change in wave characteristics (all in vacuum) from microwaves to X-rays in electromagnetic spectrum ? **1**

	Speed	**Wavelength**	**Frequency**
(A)	Remains same	Decreases	Remains same
(B)	Remains same	Decreases	Increases
(C)	Increases	Increases	Decreases
(D)	Remains same	Increases	Remains same

CBSE 2025

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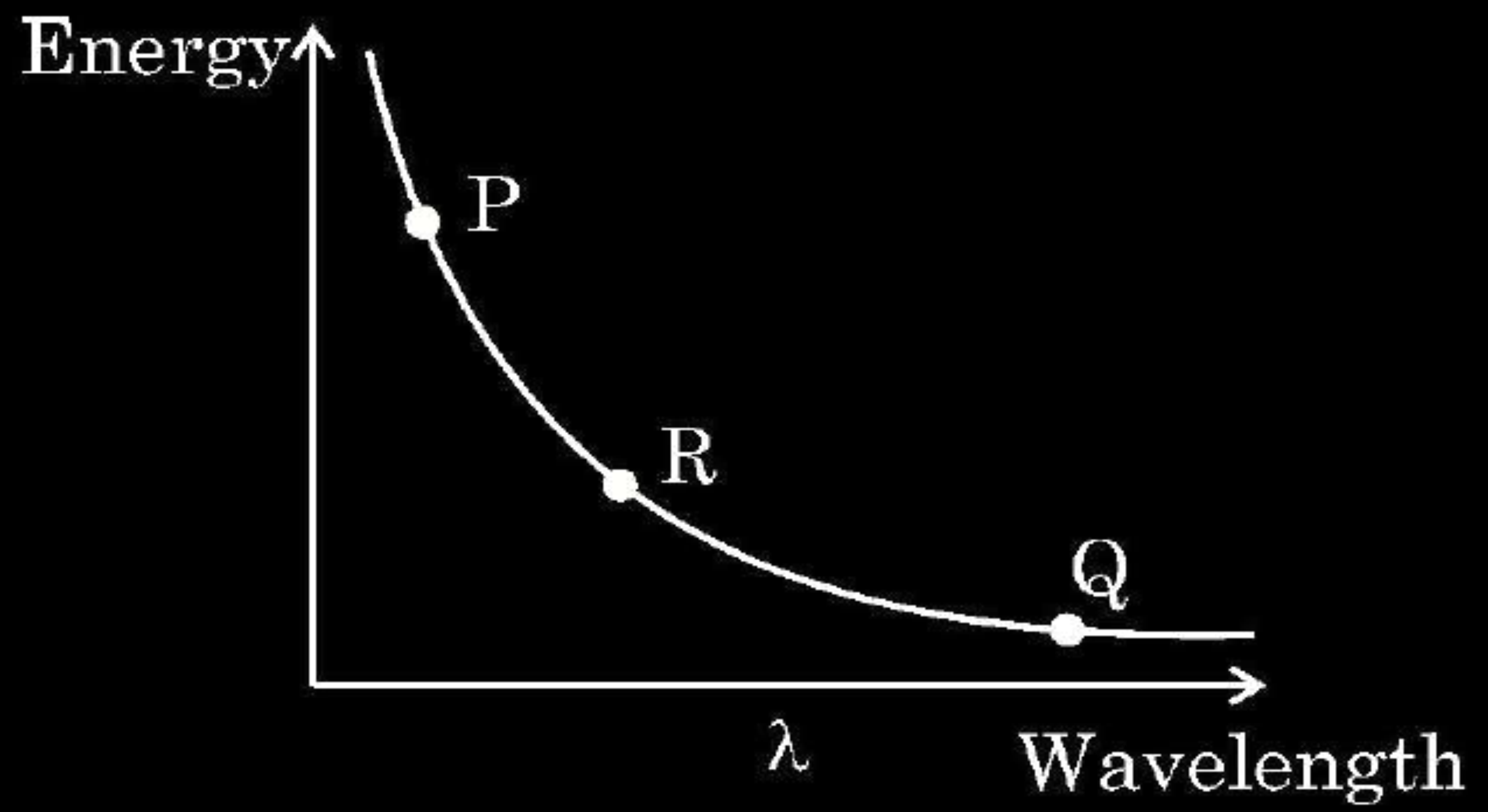
CBSE 2025

	Speed	Wavelength	Frequency
(B)	Remains same	Decreases	Increases

7. The given diagram exhibits the relationship between the wavelength of the electromagnetic waves and the energy of photon associated with them. The three points P, Q and R marked on the diagram may correspond respectively to : **1**



(A) X-rays, microwaves, UV radiation

(B) X-rays, UV radiation, microwaves

(C) UV radiation, microwaves, X-rays

(D) Microwaves, UV radiation, X-rays

CBSE 2025

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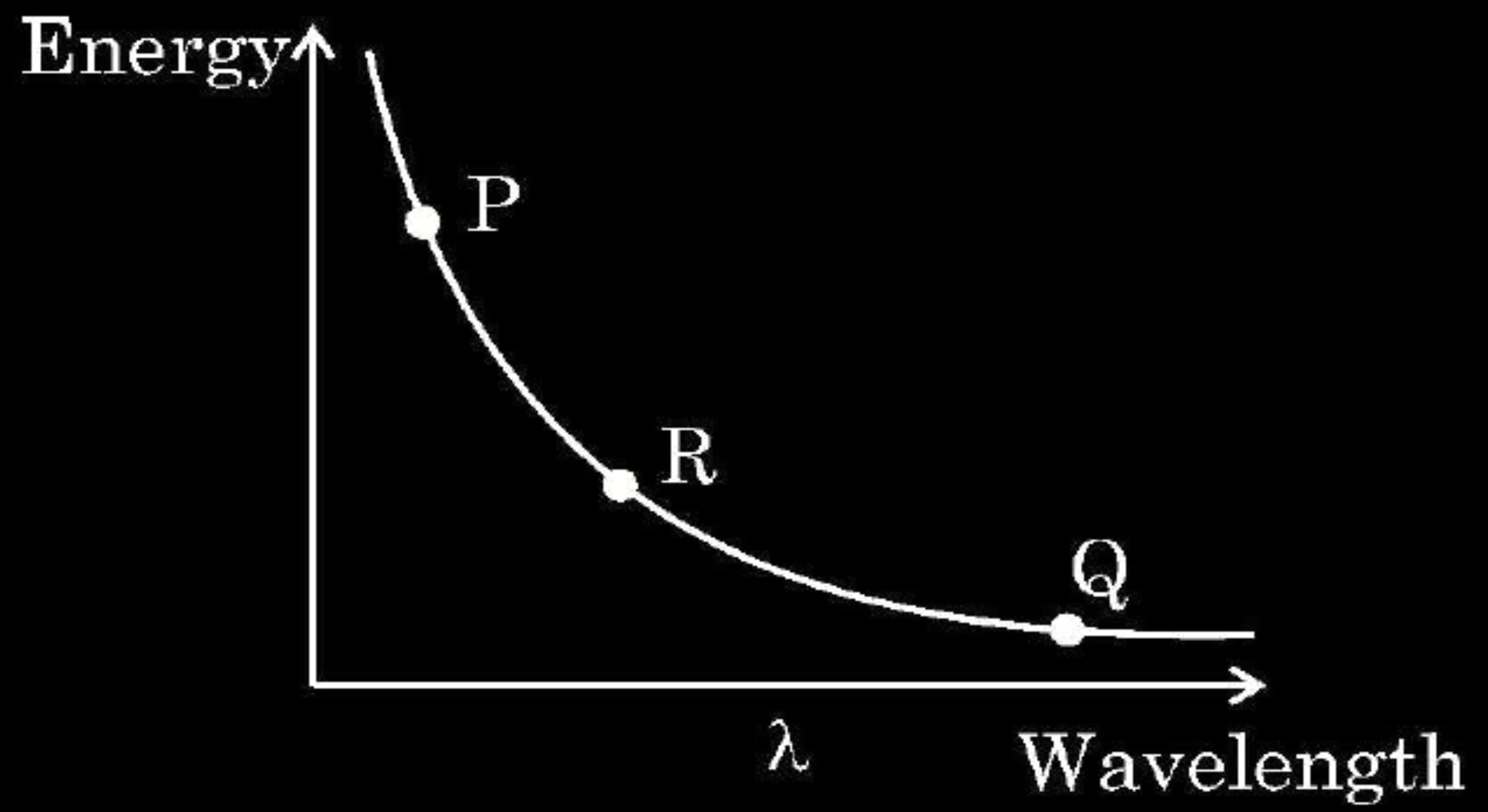

(A) X-rays, microwaves, UV radiation
(B) X-rays, UV radiation, microwaves
(C) UV radiation, microwaves, X-rays
(D) Microwaves, UV radiation, X-rays

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(A) X- rays , Micro waves , UV radiation

1. The electric field E associated with an electromagnetic wave is represented by

$$E_y = E_0 \sin(kx - \omega t)$$

Which of the following statements is correct ?

(A) The wave is propagating along +x-axis.

(B) The wave is propagating along +z-axis.

(C) The magnetic field $\vec{B}$ of the wave is acting along +y-axis.

(D) The magnetic field $\vec{B}$ of the wave is acting along –x-axis.

CBSE 2024 C

1. The electric field E associated with an electromagnetic wave is represented by

$$E_y = E_0 \sin(kx - \omega t)$$

Which of the following statements is correct ?

(A) The wave is propagating along +x-axis.

(B) The wave is propagating along +z-axis.

(C) The magnetic field $\vec{B}$ of the wave is acting along +y-axis.

(D) The magnetic field $\vec{B}$ of the wave is acting along –x-axis.

CBSE 2024 C

(A) The wave is propagating along +x-axis.

24. Explain the following, giving proper reason : *3*

(a) During charging of a capacitor, displacement current exists in the capacitor. But there is no displacement current when it gets fully charged.

(b) The frequency of microwaves in ovens matches with the resonant frequency of water molecules.

(c) Infrared waves are also known as heat waves.

CBSE 2024 C

24. Explain the following, giving proper reason : *3*

(a) During charging of a capacitor, displacement current exists in the capacitor. But there is no displacement current when it gets fully charged.

(b) The frequency of microwaves in ovens matches with the resonant frequency of water molecules.

(c) Infrared waves are also known as heat waves.

CBSE 2024 C

(a) $i_d = \varepsilon_0 \dfrac{d\varphi_E}{dt}$ Displacement current is due to changing electric flux. During charging of capacitor, there is change in electric flux. When fully charged the electric field does not change hence electric flux also does not change. Hence no displacement current.	1
(b) The frequency of the microwaves is selected to match the resonant frequency of water molecules so that energy from the waves is transferred efficiently to the kinetic energy of the molecules.	1
(c) Infrared waves are also known as heat waves, because water molecules present in most materials readily absorbs infrared waves and their thermal motion increases and heat up.	1

6. The magnetic field of an electromagnetic wave is represented as $B_x = B_0 \sin(ky - \omega t)$. It means that the wave propagation direction and wave vector k are respectively : CBSE 2024 C

(A) $+$ z axis, $\frac{2\pi}{v}$

(B) $-$ z axis, $\frac{2\pi}{T}$

(C) $+$ y axis, $\frac{2\pi}{\lambda}$

(D) $-$ y axis, $\frac{\lambda}{2\pi}$

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CBSE 2024 C

(A) + z axis, $\frac{2\pi}{v}$

(B) – z axis, $\frac{2\pi}{T}$

(C) + y axis, $\frac{2\pi}{\lambda}$

(D) – y axis, $\frac{\lambda}{2\pi}$

(C) +y axis, $\frac{2\pi}{\lambda}$

7. The speed of electromagnetic waves in a medium depends upon :

(A) properties of the medium

(B) frequency of the waves

(C) wavelength of the waves

(D) intensity of the waves

CBSE 2024 C

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(A) properties of the medium

(B) frequency of the waves

(C) wavelength of the waves

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CBSE 2024 C

(A) properties of the medium

9. The electromagnetic waves used to purify water are

(A) Infrared rays

(B) Ultraviolet rays

(C) X-rays

(D) Gamma rays

CBSE 2024

9. The electromagnetic waves used to purify water are

(A) Infrared rays (B) Ultraviolet rays

(C) X-rays (D) Gamma rays

CBSE 2024

(B) Ultraviolet rays

24. (a) Name the parts of the electromagnetic spectrum which are (i) also known as 'heat waves' and (ii) absorbed by ozone layer in the atmosphere.

(b) Write briefly one method each, of the production and detection of these radiations.

CBSE 2024

24. (a) Name the parts of the electromagnetic spectrum which are (i) also known as 'heat waves' and (ii) absorbed by ozone layer in the atmosphere.

(b) Write briefly one method each, of the production and detection of these radiations.

CBSE 2024

(a) (i) Infrared waves	½
(ii) Ultraviolet Rays	½
(b) Method of production	
Infrared waves: Hot bodies / Vibration of atoms and molecules	½
Ultraviolet Rays: Special UV lamps / Sun / Very hot bodies	½
Method of detection	
Infrared waves: Thermopiles / IR photographic film / Bolometer	½
Ultraviolet Rays: Photocells / photographic film	½

25. Name the part of the electromagnetic spectrum which are

(i) stopped by face mask worn by welders.

(ii) used in detectors in Earth satellites.

(iii) used in 'short-wave band' in communication.

Also write the order of wavelengths, in each case. CBSE 2024

25. Name the part of the electromagnetic spectrum which are

(i) stopped by face mask worn by welders.

(ii) used in detectors in Earth satellites.

(iii) used in 'short-wave band' in communication.

Also write the order of wavelengths, in each case.

CBSE 2024

(i) Ultraviolet rays	½
Order of wavelength 400 nm – 1 nm	½
(ii) Infrared waves	½
Order of wavelength 1 nm – 700 nm	½
(iii) Radio waves	½
Order of wavelength > 0.1 m	½

3. The phase difference between electric field $\vec{E}$ and magnetic field $\vec{B}$ in an electromagnetic wave propagating along z-axis is – **1**

(A) zero (B) π

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

CBSE 2024

3. The phase difference between electric field $\vec{E}$ and magnetic field $\vec{B}$ in an electromagnetic wave propagating along z-axis is – **1**

(A) zero (B) π

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

CBSE 2024

(A) Zero.

27. Explain the following giving reasons :

(i) 'Electromagnetic waves differ considerably in their mode of interaction with matter'.

(ii) 'Food items to be heated in microwave oven must contain water'.

(iii) 'Welders wear face mask with glasses during welding'. CBSE 2024 **3**

27. Explain the following giving reasons :

(i) 'Electromagnetic waves differ considerably in their mode of interaction with matter'.

(ii) 'Food items to be heated in microwave oven must contain water'.

(iii) 'Welders wear face mask with glasses during welding'. CBSE 2024 **3**

(i) Since they have different wavelenghts and frequencies, they differ considerably in their mode of interaction with matter.	1
(ii) Frequency of microwave matches with the resonant frequency of water molecules so that energy from wave is transferred to water molecules.	1
(iii) To protect their eyes from large amount of ultraviolet rays produced by welding arcs.	1

27. Name the electromagnetic waves with their wavelength range which are used for

(i) FM radio broadcast

(ii) detection of fracture in bones

(iii) treatment of muscular strain CBSE 2024 **3**

27. Name the electromagnetic waves with their wavelength range which are used for

(i) FM radio broadcast

(ii) detection of fracture in bones

(iii) treatment of muscular strain CBSE 2024 **3**

	Electromagnetic waves	wavelength range	
(i)	Radio waves	> 0.1 m	½ + ½
(ii)	X- rays	$1\text{nm} - 10^{-3}$ nm	½ + ½
(iii)	Infrared waves	1 mm - 700 nm	½ + ½

8. A plane electromagnetic wave is travelling in air in + x direction. At a particular moment, its electric field $\vec{E}$ is along + y direction. At that moment, the magnetic field $\vec{B}$ is along :

(A) + z direction and in phase with $\vec{E}$

(B) – z direction and in phase with $\vec{E}$

(C) + z direction and out of phase with $\vec{E}$

(D) – z direction and out of phase with $\vec{E}$

CBSE 2024

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(A) + z direction and in phase with $\vec{E}$

(B) – z direction and in phase with $\vec{E}$

(C) + z direction and out of phase with $\vec{E}$

(D) – z direction and out of phase with $\vec{E}$

CBSE 2024

(A) +z direction and in phase with $\vec{E}$

25. The electric field in an electromagnetic wave in vacuum is given by :

$$\vec{E} = (6 \cdot 3 \text{ N/C}) [\cos (1 \cdot 5 \text{ rad/m}) \, y + (4 \cdot 5 \times 10^8 \text{ rad/s}) \, t] \; \hat{i}$$

(a) Find the wavelength and frequency of the wave. CBSE 2024

(b) What is the amplitude of the magnetic field of the wave ?

(c) Write an expression for the magnetic field of this wave. *3*

(a) $k=\dfrac{2\pi}{\lambda}$	½
$\lambda=\dfrac{2\pi}{K}=\dfrac{4\pi}{3}\,\mathrm{m}=4.18\,\mathrm{m}$	½
$\omega=2\pi\upsilon$	
$\nu=\dfrac{\omega}{2\pi}=\dfrac{4.5\times10^{8}}{2\pi}\,\mathrm{Hz}$	½
$\nu=\dfrac{9}{4\pi}\times10^{8}\,\mathrm{Hz}$	½
$\nu=7.16\times10^{-1}\ \mathrm{Hz}$	
(b) $B_0=\dfrac{E_0}{c}$	
$B_0=\dfrac{6.3}{3\times10^{8}}=2.1\times10^{-8}\,\mathrm{T}$	½
(c) $\vec{B}=2.1\times10^{-8}[(\cos 1.5\,\mathrm{rad/m})\,y+(4.5\times10^{8}\,\mathrm{rad/s})\,t]\,\hat{k}\,\mathrm{T}$	½

8. Electromagnetic waves with frequency $1 \cdot 0 \times 10^{18}$ Hz are known as :

(A) Infrared rays

(B) Ultraviolet rays

(C) X-rays

(D) Gamma rays

CBSE 2024

8. Electromagnetic waves with frequency $1{\cdot}0 \times 10^{18}$ Hz are known as :

(A) Infrared rays

(B) Ultraviolet rays

(C) X-rays

(D) Gamma rays

CBSE 2024

(C) X- rays

8. Which one of the following has the highest frequency ?

(A) Infrared rays

(B) Gamma rays

(C) Radio waves

(D) Microwaves

CBSE 2024

8. Which one of the following has the highest frequency ?

(A) Infrared rays

(B) Gamma rays

(C) Radio waves

(D) Microwaves

CBSE 2024

(B) Gamma rays

7. In four regions I, II, III and IV, the magnetic field is given by :

I. $B_y = B_0 \sin kz$

II. $B_y = B_0 \cos kz$

III. $B_y = B_0 \sin (kz - \omega t)$

IV. $B_y = B_0 \sin kz + B_0 \cos kz$

The electromagnetic wave will exist in the region :

(A) IV

(B) I

(C) III

(D) II

CBSE 2024

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I. $B_y = B_0 \sin kz$

II. $B_y = B_0 \cos kz$

III. $B_y = B_0 \sin (kz - \omega t)$

IV. $B_y = B_0 \sin kz + B_0 \cos kz$

The electromagnetic wave will exist in the region :

(A) IV

(B) I

(C) III

(D) II

CBSE 2024

(C) III

23. (a) On what factors does the speed of an electromagnetic wave in a medium depend ?

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(b) How is an electromagnetic wave produced ?

(c) Sketch a schematic diagram depicting the electric and magnetic fields for an electromagnetic wave propagating along z-axis. *3*

23. (a) On what factors does the speed of an electromagnetic wave in a medium depend ?

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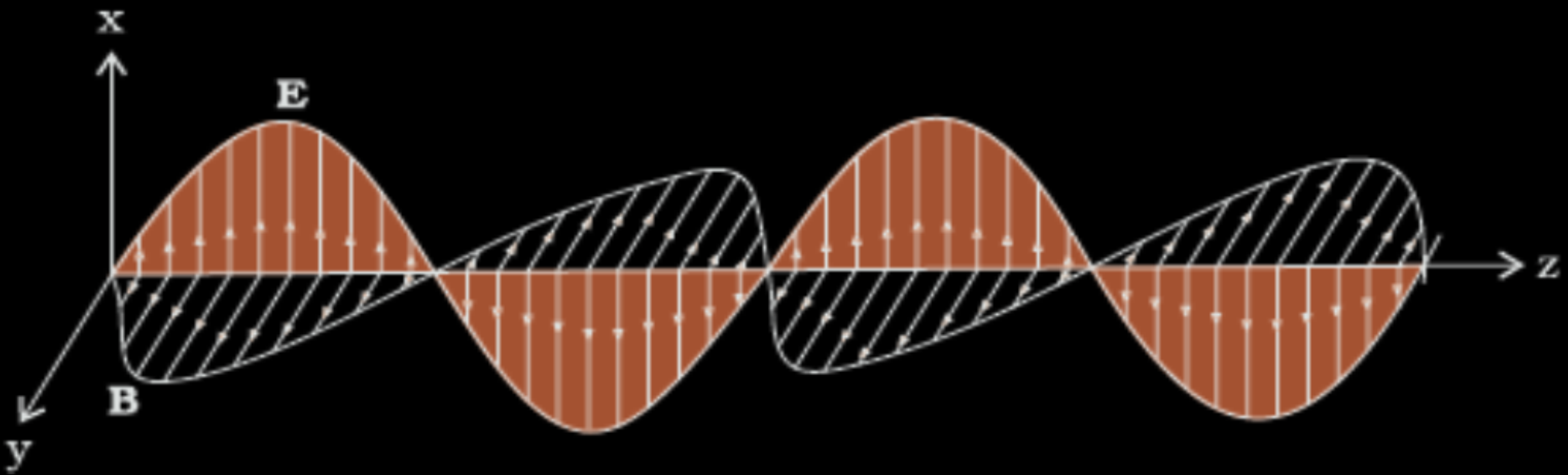
(b) How is an electromagnetic wave produced ?

(c) Sketch a schematic diagram depicting the electric and magnetic fields for an electromagnetic wave propagating along z-axis. *3*

(a) Speed of EM waves $v = \dfrac{1}{\sqrt{\mu\varepsilon}}$ Speed depends upon (i) Permittivity (ε) of medium (ii) Magnetic permeability (μ) of medium	½ + ½
(b) Accelerated charges or oscillating charges produce electromagnetic waves	**1**
(c) 	**1**

7. Welders wear special glass goggles or face masks with glass windows to protect their eyes from radiations produced by welding arcs. These radiations are :

(A) X-rays (B) Ultraviolet rays

(C) Infrared waves (D) Gamma rays

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(B) Ultraviolet rays

(C) Infrared waves

(D) Gamma rays

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(B) Ultraviolet rays

7. Electromagnetic waves with wavelength 10 nm are called :

(A) Infrared waves

(B) Ultraviolet rays

(C) Gamma rays

(D) X-rays

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7. Electromagnetic waves with wavelength 10 nm are called :

(A) Infrared waves (B) Ultraviolet rays

(C) Gamma rays (D) X-rays

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(B) Ultraviolet rays

8. The first scientist who produced and observed electromagnetic waves of wavelengths in the range 25 mm – 5 mm was :

(A) J.C. Maxwell (B) H.R. Hertz

(C) J.C. Bose (D) G. Marconi

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(C) J.C. Bose

28. (a) "The wavelength of the electromagnetic wave is often correlated with the characteristic size of the system that radiates." Give two examples to justify this statement.

(b) (i) Long distance radio broadcasts use short-wave bands. Why ?

(ii) Optical and radio telescopes are built on the ground, but X-ray astronomy is possible only from satellites orbiting the Earth. Why ? CBSE 2024 *3*

28. (a) "The wavelength of the electromagnetic wave is often correlated with the characteristic size of the system that radiates." Give two examples to justify this statement.

(b) (i) Long distance radio broadcasts use short-wave bands. Why ?

(ii) Optical and radio telescopes are built on the ground, but X-ray astronomy is possible only from satellites orbiting the Earth. Why ? CBSE 2024 *3*

a) (Any Two) • Gamma radiation having wavelength of 10^{-14} m to 10^{-15} m, typically originate from an atomic nucleus. • X-rays are emitted from heavy atoms. • Radio waves are produced by accelerating electrons in a circuit. A transmitting antenna can most efficiently radiate waves having a wavelength of about the same size as the antenna.	½ + ½
b) (i) Ionosphere reflects waves in these bands	1
(ii) Atmosphere absorbs x-rays, while visible and radio waves can penetrate it	1
Note: Full credit to be given for part (b) for mere attempt.	

8. The electric and magnetic fields of electromagnetic waves are :

(A) In the same phase and perpendicular to each other.

(B) In the same phase and not perpendicular to each other.

(C) Not in the same phase but are perpendicular to each other.

(D) Neither in the same phase nor perpendicular to each other.

CBSE 2024

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(A) In the same phase and perpendicular to each other

8. In the four regions, I, II, III and IV, the electric fields are described as :

Region I : $E_x = E_0 \sin (kz - \omega t)$

Region II : $E_x = E_0$

Region III : $E_x = E_0 \sin kz$

Region IV : $E_x = E_0 \cos kz$

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The displacement current will exist in the region :

(A) I (B) IV (C) II (D) III

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CBSE 2024

The displacement current will exist in the region :

(A) I (B) IV (C) II (D) III

(A) I

7. The frequencies of ultraviolet rays, gamma-rays and microwaves are v_1, v_2 and v_3 respectively. Then :

(A) $v_1 = v_2 = v_3$

(B) $v_1 > v_2 > v_3$

(C) $v_2 > v_1 > v_3$

(D) $v_3 > v_2 > v_1$

CBSE 2024

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CBSE 2024

(C) $v_2 > v_1 > v_3$

25. What is meant by 'displacement current' ? Show that during charging of a capacitor, displacement current is equal to conduction current in the circuit. CBSE 2024 *3*

25. What is meant by 'displacement current' ? Show that during charging of a capacitor, displacement current is equal to conduction current in the circuit. CBSE 2024 *3*

The current produced due to time varying electric field is called displacement current.	1
Electric field between the plates of a capacitor $E = \frac{\sigma}{\varepsilon_0} = \frac{q}{A\varepsilon_0}$	½
Electric flux $\phi = EA$	½
Displacement current $I_D = \varepsilon_0 \frac{d\phi}{dt}$ $= \varepsilon_0 \frac{d}{dt}\left(\frac{q}{\varepsilon_0}\right)$ $= \frac{dq}{dt}$	½
$I_D = I_C$	½

7. The speed of electromagnetic waves in vacuum is :

(a) $\dfrac{1}{\sqrt{\mu_0 \varepsilon_0}}$

(b) $\sqrt{\mu_0 \varepsilon_0}$

(c) $\mu_0 \varepsilon_0$

(d) $\dfrac{1}{\mu_0 \varepsilon_0}$

CBSE 2023 C

7. The speed of electromagnetic waves in vacuum is :

(a) $\frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ (b) $\sqrt{\mu_0 \varepsilon_0}$

(c) $\mu_0 \varepsilon_0$ (d) $\frac{1}{\mu_0 \varepsilon_0}$

CBSE 2023 C

(a) $\frac{1}{\sqrt{\mu_o \varepsilon_o}}$

20. (a) Identify the part of electromagnetic spectrum which is : *2*

(i) suitable for radar systems.

(ii) sometimes referred to as 'heat waves'.

Write their wavelength range. CBSE 2023 C

20. (a) Identify the part of electromagnetic spectrum which is : *2*

(i) suitable for radar systems.

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Write their wavelength range. CBSE 2023 C

(i) Microwave	½
Wavelength range (1mm-0.1m)	½
(ii) Infrared wave	½
Wavelength range (700nm-1mm)	½

(b) Write two characteristics of electromagnetic waves. Name the radiation used to kill germs in water purifiers. Write the range of their frequency. CBSE 2023 C *2*

(b) Write two characteristics of electromagnetic waves. Name the radiation used to kill germs in water purifiers. Write the range of their frequency. CBSE 2023 C **2**

1. EMW travel with speed of light in vaccum. 2. EMW carries energy and momentum. 3. Speed of EMW in vaccum is given by $c = \frac{1}{\sqrt{\mu_o \varepsilon_0}}$ 4. EMW are transverse in nature 5.In EMW electric and magnetic field are perpendicular to each other and to the direct propagation. (any two) Not: Award full marks, if a student write any other characteristics.	½ + ½
Ultraviolet(UV)	½
Frequency range 10^{15}-10^{17} Hz	½

5. Which of the following radiations has the highest frequency ?

(a) Visible light

(b) Infrared rays

(c) Microwaves

(d) X-rays

CBSE 2023 C

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(a) Visible light

(b) Infrared rays

(c) Microwaves

(d) X-rays

CBSE 2023 C

(d) x- rays

19. (a) The electric field of an electromagnetic wave passing through vacuum is represented as $E_x = E_0 \sin(kz - \omega t)$. Identify the parameter which is related to the (i) wavelength, and (ii) the frequency of the wave in the above equation.

(b) Write two properties of a medium that determine the velocity of light in that medium. CBSE 2023 C *2*

19. (a) The electric field of an electromagnetic wave passing through vacuum is represented as $E_x = E_0 \sin (kz - \omega t)$. Identify the parameter which is related to the (i) wavelength, and (ii) the frequency of the wave in the above equation.

(b) Write two properties of a medium that determine the velocity of light in that medium. CBSE 2023 C *2*

(a) Parameter relating wavelength is 'k'$\left(=\dfrac{2\pi}{\lambda}\right)$	½
Parameter relating frequency is ' ω ' $(= 2\pi\nu)$	½
(b) 1. Electric properties of the medium	½
2. Magnetic properties of the medium	½
Alternatively:	
i. Permittivity (ε) of the medium	
ii. Permeability (μ) of the medium	

5. Which of the following physical quantities remain the same for X-ray, red light and radio waves when travelling through a medium ?

(a) Wavelength (b) Speed

(c) Frequency (d) Momentum

CBSE 2023 C

5. Which of the following physical quantities remain the same for X-ray, red light and radio waves when travelling through a medium ?

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CBSE 2023 C

(b) speed

9. In a plane electromagnetic wave, the magnetic field oscillates sinusoidally with a frequency of 3×10^{10} Hz, and amplitude $2{\cdot}4 \times 10^{-8}$ T. The amplitude of the oscillating electric field is :

(a) $1{\cdot}6 \text{ Vm}^{-1}$ (b) $3{\cdot}2 \text{ Vm}^{-1}$

(c) $7{\cdot}2 \text{ Vm}^{-1}$ (d) 8 Vm^{-1}

CBSE 2023

9. In a plane electromagnetic wave, the magnetic field oscillates sinusoidally with a frequency of 3×10^{10} Hz, and amplitude $2{\cdot}4 \times 10^{-8}$ T. The amplitude of the oscillating electric field is :

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(c) $7{\cdot}2\ \text{Vm}^{-1}$ (d) $8\ \text{Vm}^{-1}$

CBSE 2023

(c) $7.2\ \text{V}\,m^{-1}$

20. (a) What is meant by displacement current ? Explain. Give an example where such current exists. CBSE 2023 *2*

20. (a) What is meant by displacement current ? Explain. Give an example where such current exists. CBSE 2023 *2*

The current due to change in electric field / flux is called displacement current.	½
During charging and discharging of a capacitor the electric field between the plates of the capacitor changes with time which results in displacement current. $$I_d = \varepsilon_o \frac{d\phi_E}{dt}$$	1
Displacement current is produced between the plates of a capacitor during it charging and discharging of capacitor.	½

(b) Write the wavelength range of infrared waves. Why are these waves often called 'heat waves' ? Explain. $\boxed{\text{CBSE 2023}}$ *2*

(b) Write the wavelength range of infrared waves. Why are these waves often called 'heat waves' ? Explain. CBSE 2023 *2*

Range of infrared wave is 1 mm to 700 nm.	½
Because water molecules present in most of the materials readily absorb infrared waves. After absorption their thermal motion increases. They heat up and heat their surroundings.	1½

8. Name the electromagnetic waves also known as 'heat waves'.

(a) Radio waves

(b) Microwaves

(c) X-rays

(d) Infrared waves

CBSE 2023

8. Name the electromagnetic waves also known as 'heat waves'.

(a) Radio waves (b) Microwaves

(c) X-rays (d) Infrared waves CBSE 2023

(d) Infrared

25. (a) What is a displacement current ? How is it different from a conduction current ? *2*

CBSE 2023

25. (a) What is a displacement current ? How is it different from a conduction current ? *2*

CBSE 2023

Displacement current is the current produced due to changing electric field/ electric flux in a region. **Alternatively** $i_d = \epsilon_0 \dfrac{d\phi_E}{dt}$ & $I = \dfrac{dq}{dt}$	1
Difference: Current carried by a conductor due to flow of charges is called conduction current. Displacement current is not due to flow of charges but due to changing electric field/electric flux.	1

25. (a) What is a displacement current ? How is it different from a conduction current ? *2*

CBSE 2023

Displacement current is the current produced due to changing electric field/ electric flux in a region.	1
Alternatively	
$i_d = \in_0 \dfrac{d\phi_E}{dt} \quad \& \; I = \dfrac{dq}{dt}$	
Difference: Current carried by a conductor due to flow of charges is called conduction current. Displacement current is not due to flow of charges but due to changing electric field/electric flux.	1

(b) Write any two characteristics of an electromagnetic wave. Why are microwaves used in radar systems ? CBSE 2023 2

(b) Write any two characteristics of an electromagnetic wave. Why are microwaves used in radar systems ? CBSE 2023 *2*

Any two characteristics: 1) No medium is required for their propagation. 2) Transverse in nature. 3) Consist of Electric and Magnetic field perpendicular to each other. 4) Energy is equally shared by electrical and magnetic field. 5) Travel with speed of light in vacuum.	½ + ½
Reason: Short wavelength, do not diffract/ unidirectional property.	1

2. The electromagnetic waves used in radar systems are :

(a) Infrared waves

(b) Ultraviolet rays

(c) Microwaves

(d) X-rays

CBSE 2023

2. The electromagnetic waves used in radar systems are :

(a) Infrared waves (b) Ultraviolet rays

(c) Microwaves (d) X-rays

CBSE 2023

(c) Microwaves

5. A capacitor is charged by a battery. Between the plates, during process of charging :

(a) only displacement current exists.

(b) only conduction current exists.

(c) both displacement current and conduction current exist.

(d) no current exists.

CBSE 2023

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CBSE 2023

(a) Only displacement current exists.

21. What is meant by the term 'displacement current' ? Briefly explain how this current is different from a conduction current. CBSE 2023 *2*

21. What is meant by the term 'displacement current' ? Briefly explain how this current is different from a conduction current. CBSE 2023 *2*

Displacement current: It is the current that arises due to the rate of change of electric field/flux. **Alternatively:-** $I_d = \varepsilon_0 \left(\frac{d\phi_E}{dt} \right)$ **Alternatively:** The term with units of current to explain the continuity of current in a region.	1
Difference: Displacement current is due to change in electric flux. Conduction current is due to flow of electrons. **Alternatively:** $I_d = \varepsilon_0 \left(\frac{d\phi_E}{dt} \right)$ $I_c = \frac{dq}{dt}$	1

12. Electromagnetic waves of wavelength of the order of a few meters were first produced and detected in the laboratory by :

(a) J.C. Maxwell

(b) J.C. Bose

CBSE 2023

(c) H.R. Hertz

(d) G. Marconi

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CBSE 2023

(c) H.R. Hertz

7. The electromagnetic radiations used to kill germs in water purifiers are called :

(a) Infrared waves

(b) X-rays

(c) Gamma rays

(d) Ultraviolet rays

CBSE 2023

7. The electromagnetic radiations used to kill germs in water purifiers are called :

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(b) X-rays

(c) Gamma rays

(d) Ultraviolet rays

CBSE 2023

(d) ultraviolet

5. In the process of charging of a capacitor, the current produced between the plates of the capacitor is :

(a) $\mu_0 \dfrac{d\phi_E}{dt}$

(b) $\dfrac{1}{\mu_0} \dfrac{d\phi_E}{dt}$

(c) $\varepsilon_0 \dfrac{d\phi_E}{dt}$

(d) $\dfrac{1}{\varepsilon_0} \dfrac{d\phi_E}{dt}$

where symbols have their usual meanings.

CBSE 2023

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(d) $\frac{1}{\varepsilon_0} \frac{d\phi_E}{dt}$

where symbols have their usual meanings.

CBSE 2023

(c) $\in_o \frac{d\phi_E}{dt}$

24. Identify the part of electromagnetic spectrum which is :

(a) next to the lowest frequency end of the visible part of electromagnetic spectrum, and

(b) produced by bombarding a metal target by high energy electrons.

Give one use of each of them. CBSE 2023 *2*

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(a) next to the lowest frequency end of the visible part of electromagnetic spectrum, and

(b) produced by bombarding a metal target by high energy electrons.

Give one use of each of them. CBSE 2023 *2*

a) Infrared Rays	½
Uses: -Muscular pain therapy	
(any one) -Remote control	
- Photography in foggy conditions	½
b) X-rays	
Uses: -To study crystal structure	½
(any one) -Detection of fracture in bones	
-Cancer treatment	½
Any other correct use.	

22. A plane electromagnetic wave propagates in vacuum along x-axis. CBSE 2023

(a) Give the direction of electric field and magnetic field vectors.

(b) E_0 and B_0 are the magnitudes of electric and magnetic fields respectively in a plane electromagnetic wave. Find the ratio of energy densities of electric field and magnetic field. *2*

22. A plane electromagnetic wave propagates in vacuum along x-axis. CBSE 2023

(a) Give the direction of electric field and magnetic field vectors.

(b) E_0 and B_0 are the magnitudes of electric and magnetic fields respectively in a plane electromagnetic wave. Find the ratio of energy densities of electric field and magnetic field. *2*

a) Electric field vector and magnetic field vector are along y-axis and z-axis or vice versa.	1
b) $u_E = \frac{1}{2}\in_o E^2{}_o \quad u_B = \frac{1}{2}\frac{B^2{}_o}{\mu_o}$ $\frac{u_E}{u_B} = \mu_o \in_o \frac{E_o{}^2}{B_o{}^2}$	½
Using $\frac{E_o}{B_o} = c$ and $c = \frac{1}{\sqrt{\mu_o \in_o}}$ $\frac{u_E}{u_B} = \mu_o \varepsilon_o c^2 = 1$	½
Note: Award full credit of 1 mark even if a student finds ratio by taking u_E and u_B as equal.	

20. Consider an induced magnetic field due to changing electric field and an induced electric field due to changing magnetic field. Which one is more easily observed ? Justify your answer. CBSE 2023 *2*

20. Consider an induced magnetic field due to changing electric field and an induced electric field due to changing magnetic field. Which one is more easily observed ? Justify your answer. CBSE 2023 *2*

Induced electric field due to changing magnetic field is easily observed.	1
Induced electric field due to changing magnetic field can be easily produced by various ways like rotating/moving a coil in magnetic field, changing the shape of coil in magnetic field, bringing bar magnet near a coil etc.	1

3. An electromagnetic wave is produced by a charge **1**

(a) moving with a constant velocity

(b) moving with a constant speed parallel to a magnetic field

(c) moving with an acceleration

(d) at rest

CBSE 2023

3. An electromagnetic wave is produced by a charge **1**

(a) moving with a constant velocity

(b) moving with a constant speed parallel to a magnetic field

(c) moving with an acceleration

(d) at rest

CBSE 2023

(c) Moving with an acceleration

25. (a) How are infrared waves produced ? Why are these waves referred to as heat waves ? Give any two uses of infrared waves. **2**

OR

(b) How are X-rays produced ? Give any two uses of these. CBSE 2023

25. (a) How are infrared waves produced ? Why are these waves referred to as heat waves ? Give any two uses of infrared waves. **2**

OR

(b) How are X-rays produced ? Give any two uses of these. CBSE 2023

Infrared waves are produced by hot bodies and vibrations of molecules.	½
They are referred as heat waves because they are readily absorbed by water molecules and increase their thermal energy and heat them.	½
Uses 1) Dehydration of fruits. 2) In greenhouse Effect. 3) In remote switches. (any other relevant two uses)	½ + ½
OR (b) Production of X- rays - 1 Two uses of X- rays - ½+½	
When fast moving electrons strike a heavy target like tungsten, X-rays are produced.	1
Two uses – 1. Used as a diagnostic tool in medicine, 2. Treatment for certain forms of cancer. 3. To study crystal structure. (Any two uses from above or other uses)	½ + ½

1. The ratio of the magnitudes of the electric field and magnetic field of a plane electromagnetic wave is **1**

(a) 1 (b) $\frac{1}{c}$

(c) c (d) $\frac{1}{c^2}$

CBSE 2023

1. The ratio of the magnitudes of the electric field and magnetic field of a plane electromagnetic wave is **1**

(a) 1 (b) $\frac{1}{c}$

(c) c (d) $\frac{1}{c^2}$

CBSE 2023

(c) c

7. $\vec{E}$ and $\vec{B}$ represent the electric and the magnetic field of an electromagnetic wave respectively. The direction of propagation of the wave is along **1**

(a) $\vec{B}$ (b) $\vec{E}$

(c) $\vec{E} \times \vec{B}$ (d) $\vec{B} \times \vec{E}$

CBSE 2023

7. $\vec{E}$ and $\vec{B}$ represent the electric and the magnetic field of an electromagnetic wave respectively. The direction of propagation of the wave is along **1**

(a) $\vec{B}$ (b) $\vec{E}$

(c) $\vec{E} \times \vec{B}$ (d) $\vec{B} \times \vec{E}$

CBSE 2023

(c)

$\vec{E} \times \vec{B}$

5. Which one of the following electromagnetic radiation has the least wavelength ? **1**

(A) Gamma rays (B) Microwaves

(C) Visible light (D) X-rays

CBSE 2023

5. Which one of the following electromagnetic radiation has the least wavelength ? **1**

(A) Gamma rays (B) Microwaves

(C) Visible light (D) X-rays

CBSE 2023

(A) Gamma rays

21. How are electromagnetic waves produced ? Write their two characteristics. **2**

CBSE 2023

1. Charged particle moving with varying speed. 2. Oscillating charge 3. By using LC circuit **(any one)**	1
Characteristics of em waves **(Any two)** (1) Travel with speed of light in vacuum (2) Transverse in nature (3) No medium required for propagation (4) Can be polarized (5) Exert pressure (Any other relevant characteristics)	½+ ½

5. Choose the correct option related to wavelengths (λ) of different parts of electromagnetic spectrum. **1**

(A) $\lambda_{\text{x-rays}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{\text{visible}}$

(B) $\lambda_{\text{visible}} > \lambda_{\text{x-rays}} > \lambda_{\text{radio waves}} > \lambda_{\text{micro waves}}$

(C) $\lambda_{\text{radio wave}} > \lambda_{\text{micro waves}} > \lambda_{\text{visible}} > \lambda_{\text{x-rays}}$

(D) $\lambda_{\text{visible}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{\text{x-rays}}$

CBSE 2023

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(C) $\lambda_{\text{radio wave}} > \lambda_{\text{micro waves}} > \lambda_{\text{visible}} > \lambda_{\text{x-rays}}$

(D) $\lambda_{\text{visible}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{\text{x-rays}}$

CBSE 2023

(C) $\lambda_{\text{radio waves}} > \lambda_{\text{microwaves}} > \lambda_{\text{visible}} > \lambda_{\text{X- rays}}$

21. Identify the electromagnetic radiation and write its wavelength range, which is used to kill germs in water purifier. Name the two sources of these radiations. CBSE 2023 **2**

21. Identify the electromagnetic radiation and write its wavelength range, which is used to kill germs in water purifier. Name the two sources of these radiations. CBSE 2023 **2**

Ultraviolet rays	½
Wavelength range 4×10^{-7}m-6×10^{-10} m	½
Sources :	
1. The Sun	½
2. Any device in which inner shell electrons in atoms move from one energy level to a lower level.	½
(Any other relevant source)	

21. Identify the electromagnetic wave whose wavelengths range is from about

(a) 10^{-12} m to about 10^{-8} m.

(b) 10^{-3} m to about 10^{-1} m. **2**

Write one use of each. CBSE 2023

21. Identify the electromagnetic wave whose wavelengths range is from about

(a) 10^{-12} m to about 10^{-8} m.

(b) 10^{-3} m to about 10^{-1} m. **2**

Write one use of each.

CBSE 2023

(a) X-rays	½
Use: used as diagnostic tool in medicine. (Any one)	
Treatment for certain forms of cancer.	½
To study crystal structure	
(Or any other one suitable use)	
Alternatively	
Gamma rays	
Use: used in medicine to destroy the cancer cell.	
(b) Microwaves	½
Use: In radar system for aircraft navigation	
(Or any other one suitable use)	½

CBSE 2022 C

11. (a) Name the electromagnetic waves which are

(i) produced during decay of a nucleus,

(ii) also known as heat waves,

(iii) absorbed by ozone layer in the atmosphere.

Write one use each of these electromagnetic waves. *3*

11. (a) Name the electromagnetic waves which are

CBSE 2022 C

(i) produced during decay of a nucleus,

(ii) also known as heat waves,

(iii) absorbed by ozone layer in the atmosphere.

Write one use each of these electromagnetic waves. *3*

(i)	γ – rays	½
	<u>Uses</u> – Radiotherapy / Sterilization and disinfection / Research purpose (Any one)	½
(ii)	Infrared Waves	½
	<u>Uses</u>- Heat sensors/Thermal imaging/ night vision equipment / Remote control (Any one)	½
(iii)	UV rays	½
	<u>Uses</u> – Water purification/ Killing of bacteria (Any one)	½

3. (a) (i) Long distance radio broadcasts use short-wave bands. Why ? *1*

(ii) The experimental demonstration of electromagnetic waves is possible only in the low frequency region (the radio wave region). Explain. CBSE 2022 C *1*

3. (a) (i) Long distance radio broadcasts use short-wave bands. Why ? *1*

(ii) The experimental demonstration of electromagnetic waves is possible only in the low frequency region (the radio wave region). Explain. CBSE 2022 C *1*

i) Ionosphere reflects waves in these bands.	1
ii) The frequency that we get even with modern electronic circuits is hardly about 10^{11} Hz, this is why experimental demonstration of electromagnetic waves had to come in low frequency region.	1

(b) (i) Draw a diagram showing a linearly polarized electromagnetic wave, propagating in the z-direction depicting $\vec{E}$ and $\vec{B}$. *1*

(ii) The wavelength of an electromagnetic wave is 10^{-10} m. Identify the wave and mention its one use. CBSE 2022 C *1*

(b) (i) Draw a diagram showing a linearly polarized electromagnetic wave, propagating in the z-direction depicting $\vec{E}$ and $\vec{B}$. *1*

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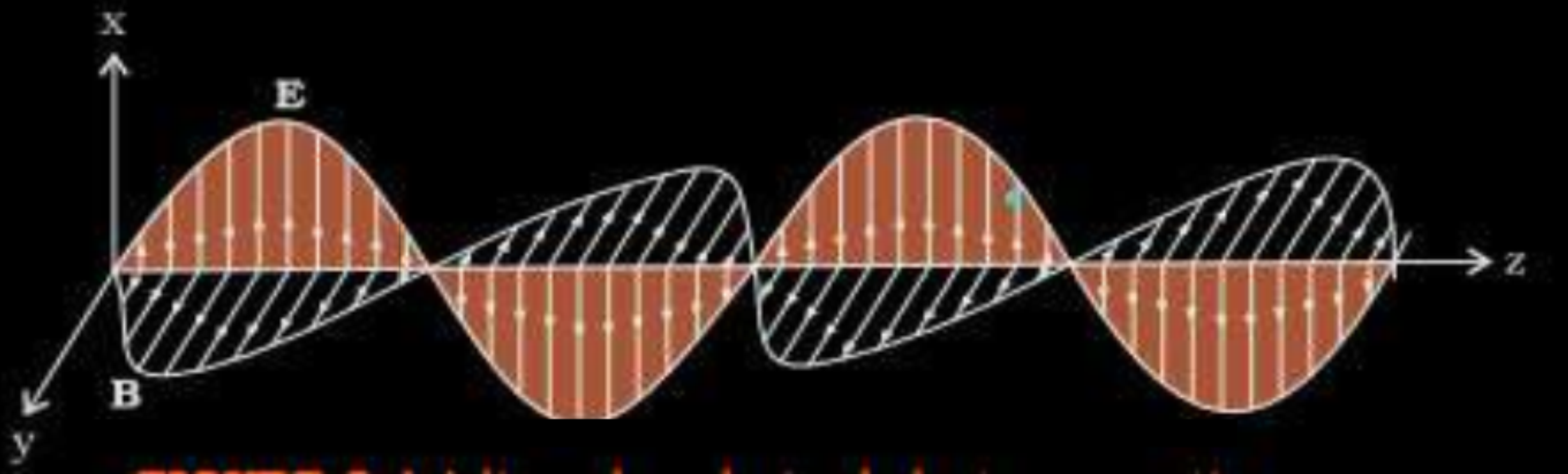
i) 	1
ii) X rays	½
Use : As a diagnostic tool in medicine / treatment of cancer / or any other.	½

9. (a) (i) Arrange the following electromagnetic radiation in the ascending order of their frequencies : 3

X-rays, microwaves, gamma rays, radio waves

(ii) Write two uses of any two of these radiation.

CBSE 2022

(i) Radio waves < microwaves < X-rays < gamma rays 1

(ii) Two uses each of any two of the radiation

Radio waves-

- TV transmission
- Radio broadcast
- Mobile communication
- Radio telescope

Microwaves-

- Microwave oven
- Speed of automobiles
- Radar
- Air craft navigation

Gamma rays-

- Treatment of cancer
- Sterilisation and disinfection

X rays-

- Diagnostic tool in medicine
- Treatment for certain forms of cancer

(Two uses of any two of these radiations) 2

7. (a) Electromagnetic waves of wavelengths λ_1, λ_2 and λ_3 are used in radar systems, in water purifiers and in remote switches of TV, respectively.

(i) Identify the electromagnetic waves, and

(ii) Write one source of each of them. CBSE 2022 **3**

7. (a) Electromagnetic waves of wavelengths λ_1, λ_2 and λ_3 are used in radar systems, in water purifiers and in remote switches of TV, respectively.

(i) Identify the electromagnetic waves, and

(ii) Write one source of each of them. CBSE 2022 **3**

	EM Waves	Sources	
$\lambda_1 \rightarrow$	Microwaves	magnetron valve / klystron valve / gun diodes	½+ ½
$\lambda_2 \rightarrow$	Ultra Violet	high voltage gas discharge tube	½+ ½
$\lambda_3 \rightarrow$	Infra red	hot bodies and molecules	½+ ½

11. (a) (i) Monochromatic light is incident on a surface separating two media. The frequency of the light after refraction remains unaffected but its wavelength changes. Why ? **3**

(ii) The frequency of an electromagnetic radiation is 1.0×10^{11} Hz. Identify the radiation and mention its two uses. CBSE 2022

11. (a) (i) Monochromatic light is incident on a surface separating two media. The frequency of the light after refraction remains unaffected but its wavelength changes. Why ? **3**

(ii) The frequency of an electromagnetic radiation is 1.0×10^{11} Hz. Identify the radiation and mention its two uses.

CBSE 2022

i) Refraction arises through interaction of incident light with the atomic constituents of matter. Atoms may be viewed as oscillators which take up the frequency of the external agency causing forced oscillations. Thus the frequency of refracted light equals the frequency of incident light. 1

Alternatively

Frequency is the characteristic of the source of light. So it remains unaffected. But λ depends on refractive index (μ) of the medium as —

$$\lambda_m = \frac{\lambda_o}{\mu}$$

ii) Infrared/ Microwaves/ Radio waves	1
Uses of Infrared rays (any two) • Remote control • Green house effect • Photography in foggy condition • To reveal secret writings • Infrared lamps **Uses of Microwaves (any two)** • Radar System • Geostationary satellite • Microwave ovens **Uses of Radiowaves (any two)** • TV transmission • Radio broadcast • Mobile transmission	½ + ½

11. (a) (i) Write the wavelength range of the following electromagnetic radiation :

Gamma-rays, Microwaves and Ultraviolet rays

(ii) Write one use each of the above mentioned radiation. CBSE 2022 *3*

11. (a) (i) Write the wavelength range of the following electromagnetic radiation :

Gamma-rays, Microwaves and Ultraviolet rays

(ii) Write one use each of the above mentioned radiation. CBSE 2022 *3*

Gamma rays – less than 10^{-11} m	½
Microwave – 10^{-1}-10^{-4} m	½
Ultraviolet rays – 10^{-7}-10^{-9} m	½
Use of gamma rays – in medicine/any other	½
Use of microwave – in radar system / heating appliances / speed gun / any other	½
Use of ultraviolet rays – UV lens / eye surgery / water purification / any other	½

CBSE 2022

10. (a) Identify electromagnetic waves which :

(i) are used in radar system.

(ii) affect a photographic plate.

(iii) are used in surgery.

Write their frequency range. *3*

10. (a) Identify electromagnetic waves which : CBSE 2022

(i) are used in radar system.

(ii) affect a photographic plate.

(iii) are used in surgery.

Write their frequency range. *3*

(i)	Microwave 10^{10}Hz – 10^{12}Hz	½ ½
(ii)	X-rays. 10^{16}Hz – 10^{20}Hz **Alternatively** Gamma Ray 10^{20}Hz – 10^{24}Hz	½ ½
(iii)	Gamma Ray 10^{20}Hz – 10^{24}Hz **Alternatively** Infrared Radiations 10^{12}Hz – 10^{14}Hz	½ ½

8. (a) (i) Depict a plane electromagnetic wave propagating along the x-axis. Write the expressions for its oscillating electric and magnetic fields.

CBSE 2022

(ii) Write three characteristics of electromagnetic waves. *3*

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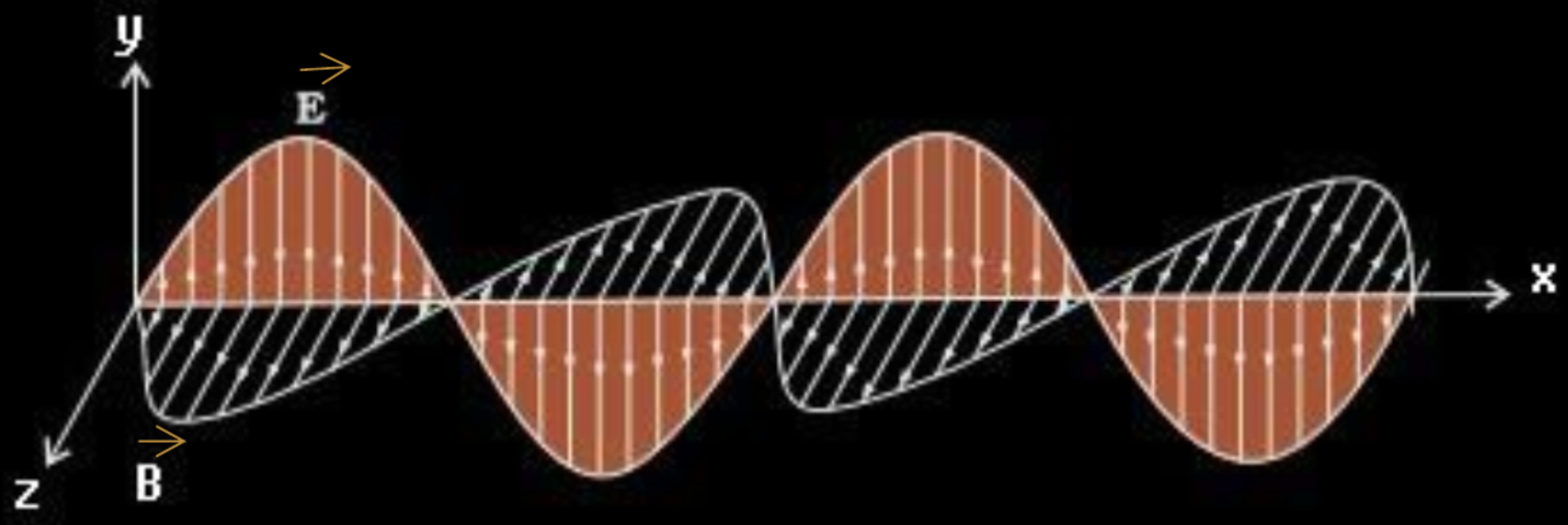
CBSE 2022

(ii) Write three characteristics of electromagnetic waves. *3*

(i)	
	½
	½
$E_y = E_0 \sin(kx - \omega t)$	½
$B_z = B_0 \sin(kx - \omega t)$	
(ii) The three characteristics are:	
a) They travel with velocity of light.	½
b) They carry energy and momentum.	½
c) They are transverse in nature.	½
(Or any other characteristic given)	

(b) Name the electromagnetic waves which are produced by the following :

(i) Radioactive decays of nucleus

(ii) Welding arcs

(iii) Hot bodies

Write one use each of these waves. CBSE 2022 *3*

(b) Name the electromagnetic waves which are produced by the following :

(i) Radioactive decays of nucleus

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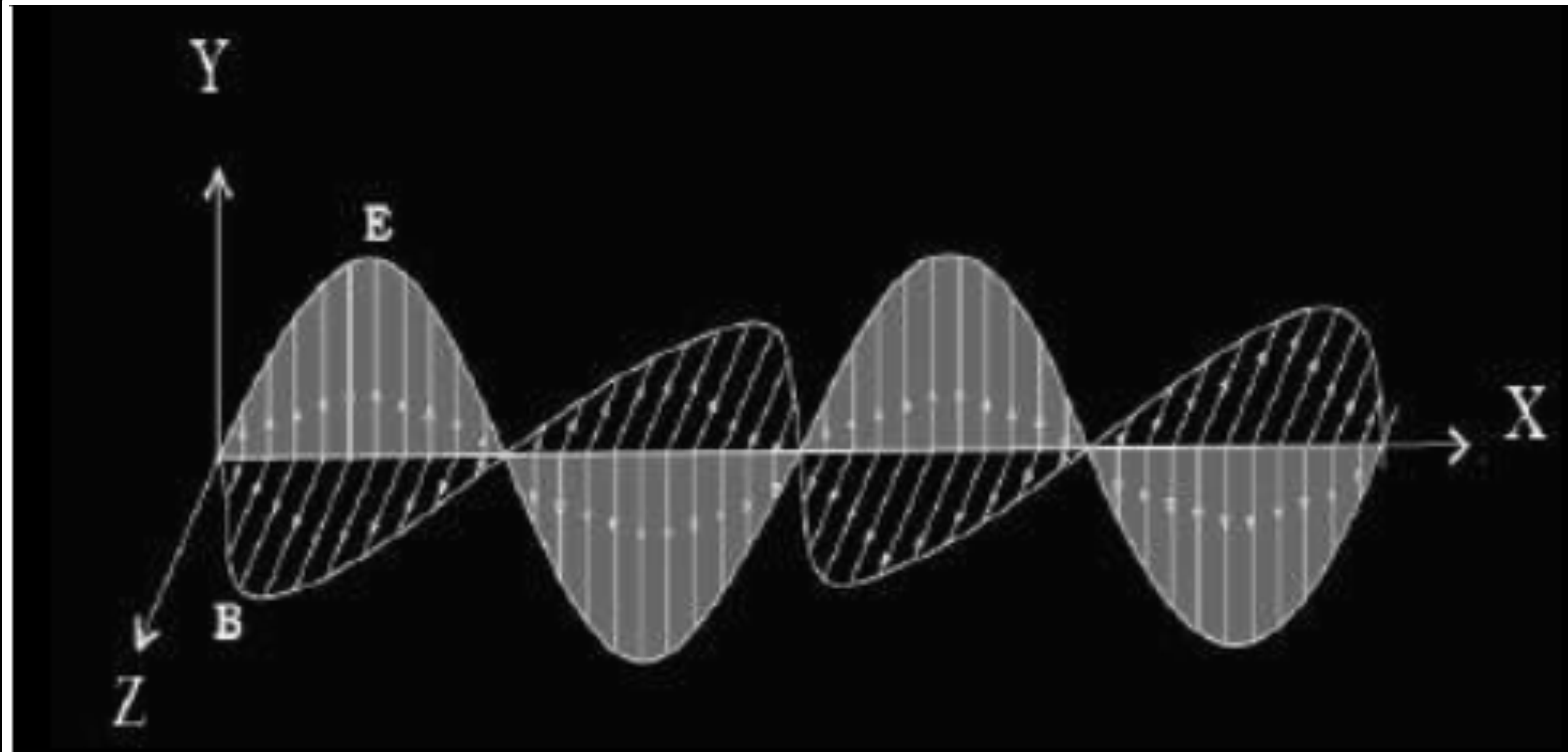
Write one use each of these waves. CBSE 2022 *3*

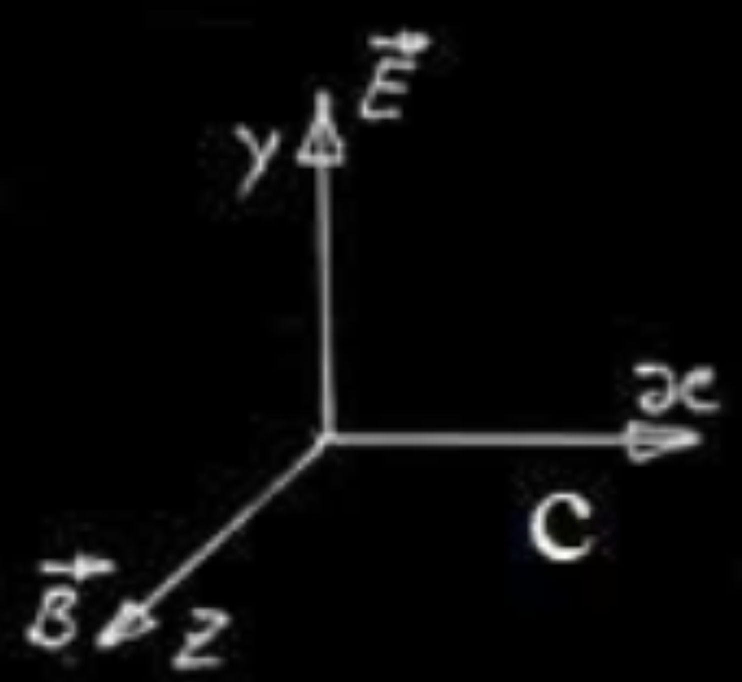
(a) Gamma Rays - Used for cancer treatment	½ + ½
(b) Ultraviolet/Visible/Infrared (either) – Use of anyone of these three.	½ + ½
(c) Infrared Rays – Used in night vision camera, bolometer & thermopiles	½ + ½
(Note: Give full credit to any other use written.)	

17. Depict the fields diagram of an electromagnetic wave propagating along positive X-axis with its electric field along Y-axis. CBSE 2020 **1**

17. Depict the fields diagram of an electromagnetic wave propagating along positive X-axis with its electric field along Y-axis. CBSE 2020 **1**



Alternatively



1

22. Gamma rays and radio waves travel with the same velocity in free space. Distinguish between them in terms of their origin and the main application. CBSE 2020 **2**

22. Gamma rays and radio waves travel with the same velocity in free space. Distinguish between them in terms of their origin and the main application. CBSE 2020 **2**

Gamma rays are emitted by radioactive nuclei/produced in nuclear reactions.	
Radio waves are produced by accelerated /oscillating charges/LC circuit.	½
Gamma rays are used for the treatment of cancer/in nuclear reactions.	½
Radio waves are used in communication systems/radio or television communication systems/cellular phones.	½
(or any other correct applications)	½

32. Name the electro-magnetic waves with their frequency range, produced in

(a) some radioactive decay

(b) sparks during electric welding

(c) TV remote

CBSE 2020

3

32. Name the electro-magnetic waves with their frequency range, produced in

(a) some radioactive decay

(b) sparks during electric welding

(c) TV remote

CBSE 2020 **3**

(a) Gamma rays, frequency range 10^{19} to 10^{24} Hz (a) UV rays, frequency range 10^{15} to 10^{17} Hz (b) Infrared rays, frequency range 10^{12} to 10^{14} Hz	½+ ½ ½+ ½ ½+ ½

19. Mention the contribution of Indian physicist J.C. Bose in the production of electromagnetic waves. CBSE 2020 *1*

19. Mention the contribution of Indian physicist J.C. Bose in the production of electromagnetic waves. CBSE 2020 *1*

J.C Bose observed / produced electromagnetic waves of short wavelength/ did very significant work in production of e.m waves.

20. Write one use of the electromagnetic waves of frequency range from 10^{16} Hz to 10^{20} Hz. CBSE 2020 *1*

20. Write one use of the electromagnetic waves of frequency range from 10^{16} Hz to 10^{20} Hz. CBSE 2020 *1*

X rays are used as diagnostic tool in medicine /
Gamma rays are used to destroy cancer cells.

24. How does an oscillating charge radiate an electromagnetic wave ? Give the relation between the frequency of radiated wave and the frequency of oscillating charge. CBSE 2020 *2*

24. How does an oscillating charge radiate an electromagnetic wave ? Give the relation between the frequency of radiated wave and the frequency of oscillating charge. CBSE 2020 *2*

An oscillating charge produces an oscillating electric field in space, which produces an oscillating magnetic field, which in turn, is a source of oscillating electric field, and so on. The oscillating electric and magnetic fields thus regenerate each other, as the wave propagates through the space.	1
The frequency of the electromagnetic wave equals the frequency of oscillation of the charge.	1

(a) Explain briefly the fact that electromagnetic waves carry energy.

(b) Why do we not feel the pressure due to sunshine ? CBSE 2020 *2*

(a) Explain briefly the fact that electromagnetic waves carry energy.

(b) Why do we not feel the pressure due to sunshine ? CBSE 2020 *2*

a) Consider a plane perpendicular to the direction of propagation of the electromagnetic wave. If there are, on this plane, electric charges, they will be set and sustained in motion by the electric and magnetic fields of the electromagnetic wave. The charges thus acquire energy and momentum from the waves.	1
b) When the sun shines on your hand, you feel the energy being absorbed from the electromagnetic waves (your hands get warm). Electromagnetic waves also transfer momentum to your hand but because c is very large, the amount of momentum transferred is extremely small and you do not feel the pressure.	1
[For any other alternative correct explanation also, award full 2 marks]	

5. Displacement current exists only when *1*

(A) electric field is changing.

(B) magnetic field is changing.

(C) electric field is not changing.

(D) magnetic field is not changing.

CBSE 2020

5. Displacement current exists only when *1*

(A) electric field is changing.

(B) magnetic field is changing.

(C) electric field is not changing.

(D) magnetic field is not changing.

CBSE 2020

(A) electric field is changing

6. Electromagnetic waves used as a diagnostic tool in medicine are *1*

(A) X-rays.

(B) ultraviolet rays.

(C) infrared radiation. CBSE 2020

(D) ultrasonic waves.

6. Electromagnetic waves used as a diagnostic tool in medicine are *1*

(A) X-rays.

(B) ultraviolet rays.

(C) infrared radiation.

(D) ultrasonic waves.

CBSE 2020

(A) X – rays

24. Which of the following electromagnetic waves has (a) minimum wavelength, and (b) minimum frequency ? Write one use of each of these two waves.

Infrared waves, Microwaves, γ-rays and X-rays $\boxed{\text{CBSE 2020}}$ *2*

24. Which of the following electromagnetic waves has (a) minimum wavelength, and (b) minimum frequency ? Write one use of each of these two waves.

Infrared waves, Microwaves, γ-rays and X-rays | CBSE 2020 | *2*

(a) minimum wavelength: γ rays	½
(b) minimum frequency: Microwaves	½
γ rays are used to treat cancer	½
Microwaves are used for communication	½
[or any other correct use]	

16. Write the mathematical form of Ampere-Maxwell circuital law. *1*

CBSE 2020

16. Write the mathematical form of Ampere-Maxwell circuital law. *1*

CBSE 2020

$$\oint B.dl = \mu_0(i_c + i_d)$$

32. (a) Write the expression for the speed of light in a material medium of relative permittivity ε_r and relative magnetic permeability μ_r.

(b) Write the wavelength range and name of the electromagnetic waves which are used in (i) radar systems for aircraft navigation, and (ii) Earth satellites to observe the growth of the crops. *3*

CBSE 2020

32. (a) Write the expression for the speed of light in a material medium of relative permittivity ε_r and relative magnetic permeability μ_r.

(b) Write the wavelength range and name of the electromagnetic waves which are used in (i) radar systems for aircraft navigation, and (ii) Earth satellites to observe the growth of the crops. *3*

CBSE 2020

(a) Speed of light in medium $$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_0\mu_r\epsilon_0\epsilon_r}}$$	1
(b) (i) Microwave range 0.1mt – 1mm $(10^{-3}m - 10^{-1}m)$	½ +½
(ii) Infrared waves range $1\ mm - 700nm$	½ + ½

Define displacement current. What role does it play while charging a capacitor by dc source. Is the value of displacement current same as that of the conduction current ? Explain. CBSE 2019

Define displacement current. What role does it play while charging a capacitor by dc source. Is the value of displacement current same as that of the conduction current ? Explain.

CBSE 2019

Displacement current is the current due to the changing of electric flux.	1
It provides continuity of current in circuits containing capacitor. $$I_d = \varepsilon_o \frac{d\phi_e}{dt}$$	1
Yes, the value of displacement is equal to the conduction current. [Explanation not required]	1

The small ozone layer on top of the stratosphere is crucial for human survival. Why ?

OR

Illustrate by giving suitable examples, how you can show that electromagnetic waves carry both energy and momentum. CBSE 2019

The small ozone layer on top of the stratosphere is crucial for human survival. Why ?

OR

Illustrate by giving suitable examples, how you can show that electromagnetic waves carry both energy and momentum. CBSE 2019

The ozone layer absorbs the UV radiations.	1
OR	
When e.m. waves falls on charged particles, they set the charges into motion. This illustrates that the e.m waves have energy and momentum. **Alternatively** When the sun shines on your hand, you feel energy being absorbed from the e.m waves **Alternatively** The radio & TV signals carry energy from one place to another (Give full marks if student explains on the basis of any one of above example) **Example** – photo electric effect	

Name the radiation of the electromagnetic spectrum which is used for the following :

(a) Radar

(b) To photograph internal parts of human body

(c) For taking photographs of the sky during night and foggy conditions

Give the frequency range in each case. CBSE 2019

Name the radiation of the electromagnetic spectrum which is used for the following :

(a) Radar

(b) To photograph internal parts of human body

(c) For taking photographs of the sky during night and foggy conditions

Give the frequency range in each case. CBSE 2019

a) Microwaves	½
$v = 10^{10}$ Hz to 10^{12} Hz	½
b) x-rays	½
$v = 10^{16}$ Hz to 10^{20} Hz	½
c) Infrared	½
$y = 10^{12}$ Hz to 10^{14} Hz	½

Which part of the electromagnetic spectrum is used in RADAR ? Give its frequency range.

CBSE 2019

OR

How are electromagnetic waves produced by accelerating charges ?

Which part of the electromagnetic spectrum is used in RADAR ? Give its frequency range.

CBSE 2019

OR

How are electromagnetic waves produced by accelerating charges ?

Microwaves Frequency range is 10^{10} to 10^{12} Hz	½ ½
OR	
• Production of electromagnetic wave 1	
Accelerated charge produces an oscillating electric field which produces an oscillating magnetic field, which is a source of oscillating electric field, and so on. Thus electromagnetic waves are produced.	1

A capacitor made of two parallel plates, each of area ‘A’ and separation ‘d’ is charged by an external dc source. Show that during charging, the displacement current inside the capacitor is the same as the current charging the capacitor. CBSE 2019

A capacitor made of two parallel plates, each of area ‘A’ and separation ‘d’ is charged by an external dc source. Show that during charging, the displacement current inside the capacitor is the same as the current charging the capacitor. CBSE 2019

A capacitor made of two parallel plates, each of area 'A' and separation 'd' is charged by an external dc source. Show that during charging, the displacement current inside the capacitor is the same as the current charging the capacitor.

CBSE 2019

$i = i_c + i_d$	1
Where i_c is conduction current and i_d is displacement current	
Outside the capacitor $i_d = 0$ so $i = i_c$	½
Inside the capacitor $i_c = 0$ so $i = i_d$	½

Mrs. Rajlakshmi had a sudden fall and was thereafter unable to stand straight. She was in great pain. Her daughter Rita took her to the doctor. The doctor took a photograph of Mrs. Rajlakshmi's bones and found that she had suffered a fracture. He advised her to rest and take the required treatment.

(a) Name the electromagnetic radiation used to take the photograph of the bones.

(b) How is this radiation produced ? CBSE 2018

(c) Mention the range of the wavelength of this electromagnetic radiation.

How is the speed of em-waves in vacuum determined by the electric and magnetic fields ?

CBSE 2017

How is the speed of em-waves in vacuum determined by the electric and magnetic fields ?

CBSE 2017

Speed of em waves is determined by the ratio of the peak values of electric and magnetic field vectors. [Alternatively, Give full credit, if student writes directly —	1

How does Ampere-Maxwell law explain the flow of current through a capacitor when it is being charged by a battery ? Write the expression for the displacement current in terms of the rate of change of electric flux.

How does Ampere-Maxwell law explain the flow of current through a capacitor when it is being charged by a battery ? Write the expression for the displacement current in terms of the rate of change of electric flux.

CBSE 2017

During charging, electric flux between the plates of capacitor keeps on changing; this results in the production of a displacement current between the plates.	1

Do electromagnetic waves carry energy and momentum ?

CBSE 2017

Do electromagnetic waves carry energy and momentum ?

CBSE 2017

Yes	**1**

Identify the electromagnetic waves whose wavelengths vary as

(a) $10^{-12}\ \text{m} < \lambda < 10^{-8}\ \text{m}$

(b) $10^{-3}\ \text{m} < \lambda < 10^{-1}\ \text{m}$

Write one use for each. CBSE 2017

Identify the electromagnetic waves whose wavelengths vary as

(a) $10^{-12}\ \text{m} < \lambda < 10^{-8}\ \text{m}$

(b) $10^{-3}\ \text{m} < \lambda < 10^{-1}\ \text{m}$

Write one use for each.

CBSE 2017

(a) X – rays	½
Used for medical purposes.	
(Also accept UV rays and gamma rays and	½
Any one use of the e.m. wave named)	
	½
(b) Microwaves	
Used in radar systems	½
(Also accept short radio waves and	
Any one use of the e.m. wave named)	

Why are microwaves considered suitable for radar systems used in aircraft navigation ? CBSE 2016

Why are microwaves considered suitable for radar systems used in aircraft navigation ? CBSE 2016

Due to their short wavelengths, (they are suitable for radar system used in aircraft navigation).	1

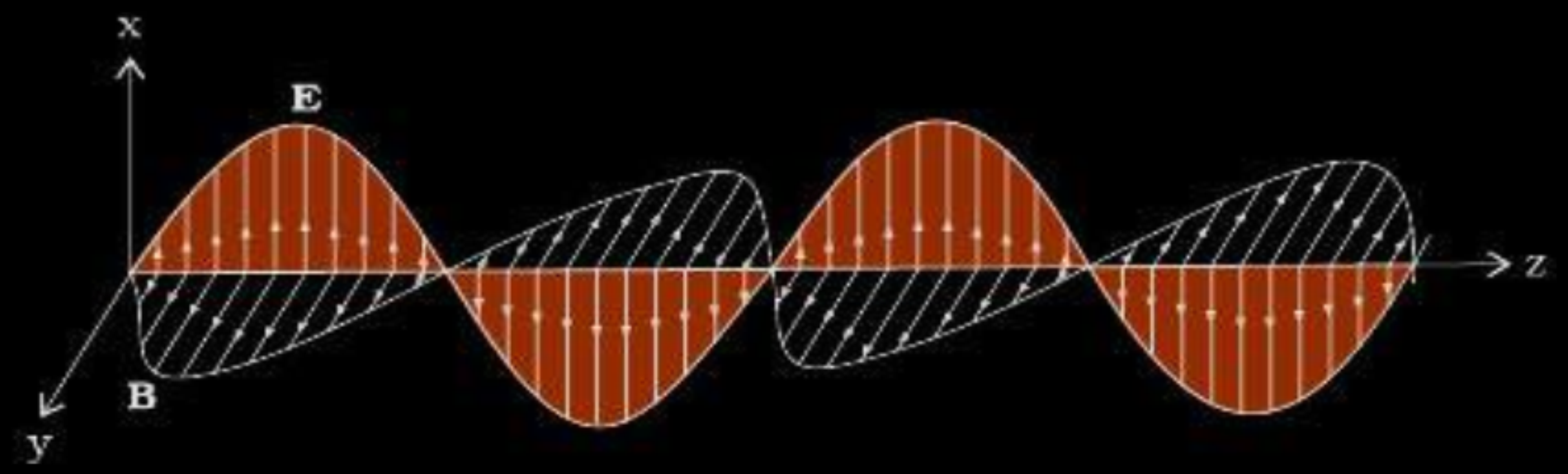
How are em waves produced by oscillating charges ?

Draw a sketch of linearly polarized em waves propagating in the Z-direction. Indicate the directions of the oscillating electric and magnetic fields. CBSE 2016

How are em waves produced by oscillating charges ?

Draw a sketch of linearly polarized em waves propagating in the Z-direction. Indicate the directions of the oscillating electric and magnetic fields.

CBSE 2016

A charge oscillating with some frequency, produces an oscillating electric field in space, which in turn produces an oscillating magnetic field perpendicular to the electric field, this process goes on repeating , producing em waves in space perpendicular to both the fields.	1
	1
Directions of $\vec{E}$ and $\vec{B}$ are perpendicular to each other and also perpendicular to direction of propagation of em waves.	½ + ½

Write Maxwell's generalization of Ampere's Circuital Law. Show that in the process of charging a capacitor, the current produced within the plates of the capacitor is

$$i = \varepsilon_0 \frac{d\Phi_E}{dt}$$

CBSE 2016

where Φ_E is the electric flux produced during charging of the capacitor plates.

Write Maxwell's generalization of Ampere's Circuital Law. Show that in the process of charging a capacitor, the current produced within the plates of the capacitor is

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CBSE 2016

where Φ_E is the electric flux produced during charging of the capacitor plates.

Ampere's circuital law is given by as $\oint \vec{B}.\vec{dl} = \mu_0 i_c$	1
But for a circuit containing capacitor, during its charging / discharging the current within the plates of the capacitor varies, (producing displacement current i_d). Therefore, the above equation, as generalized by Maxwell, is given as $\oint \vec{B}.\vec{dl} = \mu_0 i_c + \mu_0 i_d$	1
During the process of charging of capacitor, electric flux (ϕ_ϵ) between the plates of capacitor changes with time, which produces the current within the plates of capacitor. This current, being proportional to $\frac{d\phi_\epsilon}{dt}$, we have $$i = \epsilon_0 \frac{d\phi_\epsilon}{dt}$$	1

In which situation is there a displacement current but no conduction current ?

CBSE 2016

In which situation is there a displacement current but no conduction current ?

CBSE 2016

Between plates of capacitor during charging / discharging **Alternatively,** In the region of time varying electric field	1

State two properties of electromagnetic waves. How can we show that em waves carry momentum ?

CBSE 2016

State two properties of electromagnetic waves. How can we show that em waves carry momentum ?

CBSE 2016

Any two properties of electromagnetic waves Such as (a) transverse nature (b) does not get deflected by electric fields or magnetic fields (c) same speed in vacuum for all waves (d) no material medium required for propagation (e) they get refracted, diffracted and polarised / **(any two properties)**	½ + ½
Electric charges present on a plane, kept normal to the direction of propagation of an e.m. wave can be set and sustained in motion by the electric and magnetic field of the electromagnetic wave. The charges thus acquire energy and momentum from the waves.	1

How are electromagnetic waves produced ? What is the source of energy of these waves ? Write mathematical expressions for electric and magnetic fields of an electromagnetic wave propagating along the z-axis. Write any two important properties of electromagnetic waves. CBSE 2016

How are electromagnetic waves produced ? What is the source of energy of these waves ? Write mathematical expressions for electric and magnetic fields of an electromagnetic wave propagating along the z-axis. Write any two important properties of electromagnetic waves. CBSE 2016

ε.M. waves are produced by accelerated /oscillating charges. ½

Source of energy is the source that accelerates the charges ½

Expression for the electric and magnetic fields (for an e.m. wave propagating along the z – axis) can be

$$E_x = E_0 \sin(kz - wt)$$ ½

$$B_y = B_0 \sin(kz - wt)$$ ½

<u>**Properties (any two)**</u>

(i) Transverse nature

(ii) Have a definite speed (for all frequencies) in vaccum

(iii) Can be polarized

(iv) Can show the phenomenon of interference and diffraction

(v) Can transport energy from one point to another

(vi) Have oscillating electric and magnetic fields along mutually perpendicular directions

(vii) Have a momentum associated with them.

(viii) Their speed , in a medium , depends upon the values of μ and ε for that medium. ½ + ½

(i) How are infrared waves produced ? Write their one important use.

(ii) The thin Ozone layer on top of the stratosphere is crucial for human survival. Why ?

CBSE 2016

(i) How are infrared waves produced ? Write their one important use.

(ii) The thin Ozone layer on top of the stratosphere is crucial for human survival. Why ?

CBSE 2016

(i) Infrared waves are produced by hot bodies and molecules.	½
Important use(Any one) To treat muscular strains/ To reveal the secret writings on the ancient walls/ For producing dehydrated fruits/ Solar heater/ Solar cooker	½
Ozone layer protects us from harmful U-V rays	1

(i) Identify the part of the electromagnetic spectrum which is :

(a) suitable for radar system used in aircraft navigation,

(b) produced by bombarding a metal target by high speed electrons.

(ii) Why does a galvanometer show a momentary deflection at the time of charging or discharging a capacitor ? Write the necessary expression to explain this observation. CBSE 2016

(i) Identify the part of the electromagnetic spectrum which is :

(a) suitable for radar system used in aircraft navigation,

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i. a. Microwaves	1
b. X-rays	1
ii Due to conduction current in the connecting wires and a displacement current between the plates	½
$I_d = \epsilon_0 \frac{d\phi_E}{dt}$	½